



Machine learning accelerates the materials discovery

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ABSTRACT

As the big data generated by the development of modern experiments and computing technology becomes more and more accessible, the material design method based on machine learning (ML) has opened a new paradigm for materials science research. With its ability to automatically solve complex tasks, machine learning is being used as a new method to help discover the relevance of materials, understand materials' properties, and accelerate the discovery of materials. This paper first introduces the general process of machine learning in materials science. Secondly, the applications of machine learning in material properties prediction, classification and identification, auxiliary micro-scale characterization, phase transformation research and phase diagram construction, process optimization, service behavior evaluation, accelerating the development of computational simulation technology, multi-objective optimization and inverse design of materials are reviewed. Finally, we discuss the main challenges and possible solutions in machine learning, and predict the potential research directions.

1. Introduction

The traditional development process of new materials adopts the trial and error method, and the experimental steps are cumbersome. It takes 15–25 years or even longer cycles from research and development to application. The long cycle and high-cost limitations make it difficult to accelerate the development of materials [1]. In addition, more and more techniques are used for material characterization, and the corresponding data volumes and dimensions are becoming more and more complex. Traditional methods of experimental analysis that rely on manual methods sometimes fail to properly exploit the intrinsic relationship between material characteristics and properties [2]. As an alternative, the rise of computational simulation has become a powerful tool for discovering new materials. High-precision calculation methods such as DFT-based electronic structure calculation [3], Monte Carlo simulation [4], molecular dynamics simulation [5], phase field method [6] and finite element analysis [7] are widely used in new materials discovery. However, it is often applied only to specific systems, the computation of complex systems is unaffordable, and the development of theoretical methods that do not meet the requirements for quantitative characterization of material properties [8]. Therefore, it is necessary to develop new methods to guide the design and development of new

materials.

With the development of experimental research and material theory, the above-mentioned experiments and computational simulations have produced massive data. As shown in Fig. 1, when the paradigm of materials science research has gone through the three stages of experimental science, theoretical science and simulation, these big data increase the demand for the fourth paradigm of computational science. The combination of big data and artificial intelligence is called "the fourth paradigm of science" or "the fourth industrial revolution" [9]. Machine learning is one of the core technologies of materials informatics. Based on material databases, machine learning constructs models for specific material properties and quickly achieves the prediction of material properties, which is expected to accelerate the design of new materials and shorten the development cycle of materials [10]. Because of its ability to learn behavior and trends from available data without knowing the underlying physical mechanism, it has begun to play an important role in materials science.

In recent years, machine learning has become the frontier direction and hotspot of materials research [11–15]. Juan et al. [16] reviewed the progress in materials science research using ML, focusing on meaningful applications of ML in metal materials, battery materials, photovoltaic materials and metallic glass. A summary of recent advances in the

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intersection of the fields of 2D materials and ML was provided by Yin et al. [17], who discussed atomistic structure analysis, property prediction, ML-assisted preparation of 2D material and revealed their connections. Imaging analysis was performed by Ge et al. [18] using deep learning methods to identify structures and retrieve links between microstructures and performance. It also highlights recent advances in applying deep learning analysis in microscopy imaging. Liao et al. [19] summarized the work in metaheuristic-based inverse design of various materials. Furthermore, machine learning is especially suitable for the following situations: (1) Repetitive and expensive single tests (such as material life evaluation, online monitoring of material processing or high-throughput material characterization fields, in which real-time prediction and feedback are required to avoid the accumulation of erroneous data. (2) It is very expensive or difficult to use traditional methods to characterize or calculate material properties (for example, the preparation of batch test samples for precious metal alloys is expensive, and the workload of cross-scale calculations from microstructure, mesostructure and macro properties is heavy). (3) It is a very complex research phenomenon, and its explicit solution to the basic equation is unknown yet. Examples of materials science that are suited for research using machine learning methods include the following properties [20,21]: mechanical properties of composite materials, the glass transition temperature of oxide glasses, dielectric loss of polycrystalline materials over a wide frequency and temperature range, mechanical properties of composites, the service life of alloy materials, thermoelectric properties of inorganic materials, large-scale screening and optimized design of porous materials. All of these involve the multi-scale geometric structure of the material, inherent complexity, and the existence of multi-type and multi-scale defects.

Because of the increasing importance of machine learning methods in materials research, the primary purpose of this article is to review the application of machine learning in materials science, and analyze successful experiences and existing challenges. The rest of this paper is arranged as follows: Section 2 introduces the basic process of machine learning in materials science. Section 3 introduces in detail the research status of machine learning in material properties prediction, classification and identification, auxiliary micro-scale characterization, phase transformation research and phase diagram construction, process optimization, service behavior evaluation, accelerating the development of computational simulation technology, multi-objective optimization and inverse design of materials. In Section 3, the research status of machine learning in material properties prediction, classification and identification, auxiliary micro-scale characterization, phase transformation research and phase diagram construction, process optimization, service behavior evaluation, accelerating the development of computational simulation technology, multi-objective optimization and inverse design

of materials are introduced. Section 4 discusses the common problems in machine learning, and then puts forward the corresponding countermeasures. Finally, the research interests of machine learning in materials science in the future are presented in Section 5.

2. Machine learning process

The groundbreaking use of machine learning in materials science dates back to the 1990s. Rao et al. [22] proposed an artificial neural network (ANN) method to model the macroscopic mechanical behavior of ceramic matrix composites (CMC) when studying SiC/SiC ceramic composites, and the ability of ANN to learn the complex multi-parameter interaction among various microstructure parameters was proved. Subsequently, machine learning has been used to solve various topics in materials science.

Fig. 2 shows the general workflow of ML-based material discovery and design. It involves four main steps: (I) data collection and preprocessing, (II) feature engineering, (III) model selection and training and (IV) model evaluation and optimization. The first step is to obtain the appropriate target data from the open and shared database or self-generated data, and preprocess the data into the desired format. The second step is to construct the descriptor or feature set of the target data by collecting or constructing the feature related to the prediction target. The third step is to construct the mapping model of known data between descriptor and target attribute based on statistical methods, such as simple linear regression, nonlinear regression, highly complex kernel ridge regression and neural network. In the last step, the generalization error of the model is evaluated through calculation-based testing, and the model is optimized according to the evaluation results.

2.1. Data collection and preprocessing

Data can be collected from published literature or reference books, obtained from simulations and calculations, obtained in laboratory experiments through high-throughput experimental techniques, citing data from high-quality databases, or a combination of the above methods. For example, there are a variety of material property data based on experiments and simulations (for example, physics, chemistry, electronics, thermodynamics, machinery), process parameter data (casting, heat treatment, hot working), image data (electron backscattering, type of phase, phase distribution, interface), structured data (crystal structure, electron diffraction, structural evolution) and text data (literature, reference books). Of course, several of these types of data are usually coupled together. With advances in computational simulation, experimental characterization, and computer performance, more and more materials databases are available to the public. Table 1

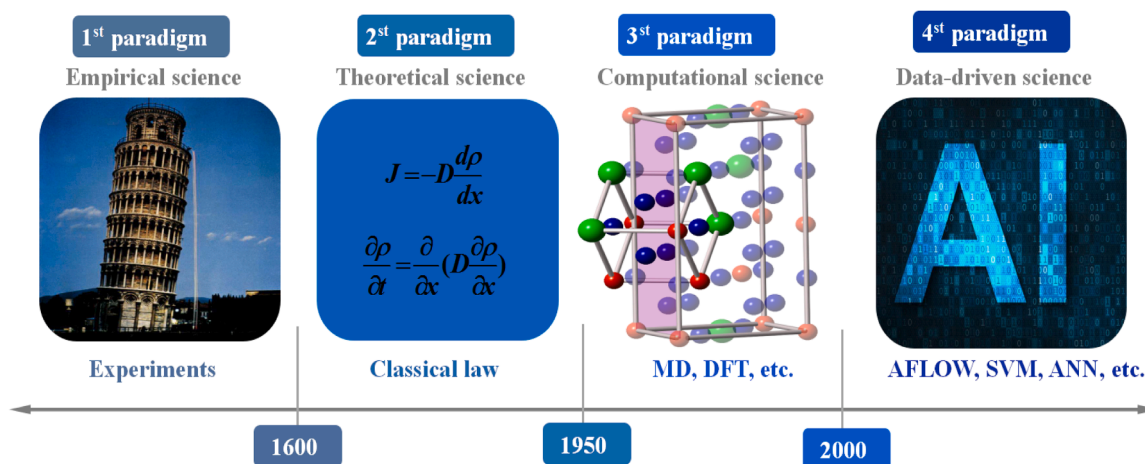


Fig. 1. The four paradigms of science: empirical, theoretical, computational and data-driven.

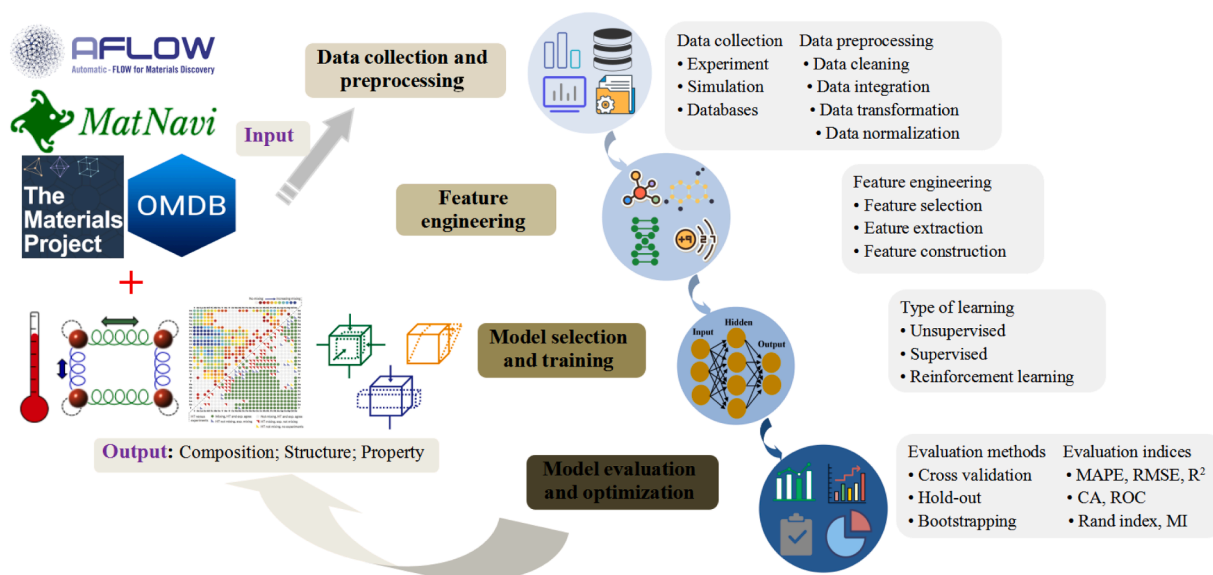


Fig. 2. The general process of machine learning in materials science.

lists several available databases containing a large number of material structures and properties. For example: AMCSDB [23], CSD [24], ChemSpider [25], COD [26], ICSD [27] and Pauling File Database [28], which are obtained based on experimental research. Moreover, CMR [29], OQMD [30], ESP, HCEP [31] and AFLOW [32] are based on theoretical simulations. Based on these material databases, learning models can be established for specific material properties to quickly predict material properties, which is expected to accelerate the design of new materials and shorten the development cycle of materials.

However, many databases still suffer from non-uniform formats (e.g., different experimental conditions or measurement techniques) or too small numbers [33]. For the former, the determination of the database standard will provide greater convenience for the application of the database. For example, the OPTIMADE consortium (<https://www.optimade.org>) is working on a common API between many different databases. For the latter problem, the reasonable shareability of commercial databases and the development of text mining technology are expected to acquire more data successfully. In addition, high-throughput/combinatorial experiments generate large amounts of experimental data, and many large open databases also have open source software platforms (e.g., Atomate [34], Pymatgen [35], FireWorks [36], AFLOW [37], and AiiDA [38]) to facilitate high-throughput computing [39]. In short, with the development of information technology and experimental technology, more ways and convenient conditions are provided for data acquisition.

The quality of the data can be improved by preprocessing the data, which can help to improve the accuracy and performance of the subsequent learning procedure. Preprocessing methods typically include data cleaning, data integration, data transformation, and data normalization, among which the most commonly used are data cleaning and data integration [40]. The four common methods are described below: (1) Data cleaning is a huge task achieved by smoothing noise, filling missing values, correcting inconsistencies and identifying outliers in data [41]. Among them, the common methods for filling in missing values are to delete samples containing missing values, fill in missing values manually, fill in missing values with a global constant, fill in missing values with the average value of attributes, fill in corresponding missing values with the average value of attributes of the same type as the given tuple, and fill in missing values with the most likely value [42]. The commonly used methods for smooth noise are binning, regression and clustering [43,44]. (2) With the emergence of big data, multi-source data integration is required, and data integration needs to consider

many issues, such as entity recognition, mainly matching real-world entities from multiple different information sources. Redundancy is another key issue. An attribute can be redundant if it can be "exported" by another attribute set. Commonly used redundant correlation analysis methods include pearson correlation coefficient [45], chi-square test [46], and covariance of numerical attributes [47]. (3) Data transformation, transforming data into a form suitable for processing according to needs, in order to discover the hidden potential patterns and laws. The commonly used data transformation methods are attribute construction, data generalization, data clustering, data normalization, data smoothing, etc. [48]. (4) The emergence of big data has made data analysis based on traditional unsupervised learning very time-consuming and complex. Data reduction techniques are used to achieve a reduced representation of a dataset, which greatly reduces the dataset size while maintaining the integrity of the original data [49]. Common data reduction methods include concept hierarchy generation, numerical reduction, subset selection of data attributes, discretization, dimension reduction, and data cube clustering [50].

2.2. Feature engineering

The dimensions of the collected data are usually relatively high, so it isn't easy to obtain the ideal effect if it is directly used to construct the machine learning prediction model. On the one hand, the higher dimensions of samples cause information redundancy, increase the amount of calculation, and even cause a "curse of dimensionality" in serious cases [51]; On the other hand, some data (noise) are target interference factors, which will adversely affect the accuracy of the model. Therefore, the primary task of constructing a machine learning model is to extract information features from the original data. Feature engineering is the process of converting raw data into features that better represent the nature of a problem, so applying these features to predictive models can reduce redundancy in modeling and enhance model interpretability [52]. Material features include many forms: (1) material composition, microstructure, structure, process parameters and other characteristics that determine material properties or service behavior; (2) physical and chemical characteristics of materials or electronic and atomic characteristics of elements; (3) According to the material theory or prior knowledge, the features in (1) and (2) above construct new features through the mathematical operation. Properly constructing target-related material features is the basis for building high-accuracy machine-learning predictive models. According to the

Table 1

An overview of some of the most authoritative databases in material science. (The database is set with hyperlinks).

Databases	Description
AFLOW	A globally available database of 3,312,125 material compounds with over 566,373,375 calculated properties, and growing.
American Mineralogist Crystal Structure Database (AMCSD)	The crystal structure database includes all structures published in Canadian Mineralogists, European Journal of Mineralogy and American Mineralogist.
Cambridge Structural Database (CSD)	A world leading comprehensive collection of small-molecule organic and organometallic crystal structures.
ChemSpider	A free database of chemical structures providing quick access to more than 67 million structures, characteristics and related information.
Citration	Experimental and computed properties of materials.
Computational Materials Repository (CMR)	Infrastructures to enable data collection, storage, retrieval and analysis from electronic structure code.
Crystallography Open Database (COD)	Open access collection of crystal structures of inorganic, organic, minerals and metal organics compounds excluding biopolymers.
The electronic structure project (ESP)	A collection of some 60,000 electronic structure calculations.
GDB Database (GDB)	The GDB-13 database is the largest publicly available database of small organic molecules to date.
Harvard Clean Energy Project	A massive database of organic solar cell materials.
HTEM	Information on materials obtained from high throughput experiments at NREL (National Renewable Energy Laboratory) is available in HTEM DB (High Throughput Experimental Materials Database).
Interatomic Potentials Repository	These repositories provide a source for interatomic potentials, related files, and evaluation tools to assist researchers in obtaining interatomic models and in judging their quality and applicability.
Khazana	Platforms for storing structural and attribute data created by atomistic simulations, and tools for designing materials through data learning.
MARVEL NCCR	The project's code, data and workflow are disseminated through the Materials Cloud Platform and the Quantum Mobile Virtual Machine, both powered by the materials' informatics framework AiiDA.
The Materials Project	Harnessing the power of supercomputing and state of the art electronic structure methods, the Materials Project provides open web-based access to computed information on known and predicted materials.
MatNavi	An integrated database system consisting of a dozen material databases such as polymer databases, inorganic materials databases, metal materials databases, and electronic structure calculation databases.
MatWeb	The MatWeb material properties database is searchable and includes data sheets of semiconductors, ceramics, fibers, thermoset and thermoplastic polymers, metals and other engineering materials.
Materials Platform for Data Science (MPDS)	Materials data provided by the MPDS, extracted from scientific publications by the project PAULING FILE team.
NIST Data Curator	The NIST Materials Data Curation System (MDCS) provides a means to capture, share, and convert material data into structured

Table 1 (continued)

Databases	Description
NIST Materials Data Repository	formats that can be converted to other formats, namely XML based formats. As part of its coordination with the Materials Genomics Research Project, the National Institute of Standards and Technology has established a Materials Science Database. NIST provides this data as a public service.
The NIST Inorganic Crystal Structure Database (ICSD)	Crystal structure data on inorganic compounds were comprehensively collected, containing more than 210,000 entries and literature from 1913.
The Novel Materials Discovery (NOMAD)	All data are provided in the original format generated by the underlying code (Repository) and in a generic, machine-processable, well-defined data format.
National Renewable Energy Laboratory Materials Database (NREL MatDB)	A database of computational materials with specific emphasis on materials for renewable energy applications including but not limited to photo-electrochemical water splitting materials, thermoelectrics and photovoltaic materials.
Open Quantum Materials Database (OQMD)	A database of DFT calculated thermodynamic and structural properties of 815654 materials.
Organic Materials Database (OMDB)	An open access electronic structure database for 3-dimensional organic crystals.
Pauling file	Crystal structure + phase diagram + physical properties of inorganic compounds in the world's largest database.
Synthesis Project	A database containing oxide synthesis parameters, open source datasets of materials science journal articles, and catalytic data.
SUNCAT	Materials informatics center for catalyst design at atomic scale. Online tools and computational results for 112,157 surface reactions and barriers available at catalysis-hub.org.
SpringerMaterials	SpringerMaterials provides curated data and advanced functionality to support research in chemistry, physics, materials science, engineering and other related areas.
The Thermoelectrics Design Laboratory (TEDesignLab)	The database contains calculated transport properties and thermoelectric performance rankings of 2701 materials.
ZINC	ZINC contains more than 230 million off-the-shelf compounds in 3D formats that are available for purchase. ZINC also contains over 750 million purchasable compounds that users can search for in less than a minute.
The Materials Cloud	Materials Cloud supports Open Science and FAIR sharing of research data in the field of Materials Science.

needs of research problems or objectives, the above features are generally used alone or in combination. For example, for application scenarios such as process parameter optimization and service behavior evaluation, the first form of material characteristics is usually directly used as model input. For example, Zhang et al. [53] studied the effects of cutting speed, feed rate and cutting depth on surface roughness of biodegradable magnesium alloy AZ31 when milling under dry conditions. Firefly and particle swarm optimization algorithm is used to optimize the surface roughness. The artificial neural network model generates the objective functions of these algorithms. If the prediction target requires extreme accuracy (such as phase structure, formation enthalpy, atomic size, or electron-phonon coupling constant), the descriptor must be fine enough. Based on the data of 322 as-cast high entropy alloys, Li et al. [54] selected five material factors such as mixing entropy, mixing enthalpy, melting temperature, atomic size difference,

and electron concentration by sequential selection method, and a support vector machine classification model was established. The classification accuracy of the model for high entropy alloy body-centered cubic single-phase, face-centered cubic single-phase, and no single-phase solid solution was more than 90%. In scenarios such as properties prediction and material composition optimization design, in order to enhance the generalization ability of the machine learning model, the third form of material features will be a better choice. For example, Zhang et al. [55] classified and identified two phases of high-entropy alloys and constructed 56 material features using 28 intrinsic physical and chemical parameters, such as effective charge, ionization energy, and melting temperature, weighted by material composition and calculated by molar averages and misalignments. In addition, 14 empirical material descriptions were obtained from the literature. A genetic algorithm is used to effectively select the best ML model and material feature combination from 70 material features and a large number of machine learning algorithms, and its effectiveness in the two-phase formation of high entropy alloy is proved. For the identification of non-solid solution and solid solution HEAs, the optimized classification model achieves 88.7% accuracy, and 91.3% accuracy for further identification of face-centered cubic, body-centered and dual phase HEAs.

People usually think of feature engineering as a problem. In fact, there are many sub-problems under feature engineering, mainly including feature selection [56], feature extraction [57] and feature construction [58]. The following three sub-questions will be introduced in detail.

Feature selection technology can simplify the complexity of the final model by reducing useless features, and the ultimate goal is to obtain a concise model that improves computational speed without reducing the accuracy of the prediction. According to the form of feature selection, the feature selection method can be divided into three categories. 1. Filter method selects features [59]. Commonly used filter methods include variance threshold [60], correlation coefficient [61], hypothesis test method [62] and mutual information [63]. The advantage of filtering feature selection is that it is more efficient in computing time, and has high robustness to overfitting problems. Disadvantages: the correlation between features is not considered, and useful correlation features may be kicked out by mistake. 2. Wrapper [64–66], commonly used wrapping methods include (1) complete search (such as breadth-first search [67], branch-and-bound search [68], beam search [69] and best-first search [70]), (2) heuristic search (such as sequence forward selection [71], sequence backward selection [72], bidirectional search [73], plus-L minus-R selection [74], sequential floating selection [75] and decision tree [76]), (3) random search (such as random generation plus sequential selection [77], simulated annealing algorithm [78], genetic algorithm [79]). Compared with the filter method, the feature subset classification performance found by the wrapper method is generally better. Disadvantages: the features selected by the wrapper method are not generic. Once the learning algorithm is modified, the feature selection must be selected again for the learning algorithms. Due to the fact that the classifier is trained and tested for each subset evaluation, the computational complexity of the algorithm is very high, especially for large datasets. 3. Embedded feature selection [80,81] (mainly including feature selection method based on penalty item and feature selection method based on tree model). The embedding method is fast and effective, but how to set parameters requires deep background knowledge.

Feature extraction is generally before feature selection. The object it extracts is the original data, and the purpose is to automatically construct new features and convert the original data into a set of features with obvious physical or statistical significance. Commonly used methods include dimensionality reduction (independent component analysis [82], linear discriminant analysis [83], principal component analysis [84], etc.), scale-invariant feature transform [85], Gabor [86], histogram of oriented gradient [87], convolutional neural network [88] in image, bag-of-words model [89], word embedding model [90] in text.

Both feature extraction and feature selection aim to find the most effective feature from the original. What distinguishes them is that feature extraction focuses on obtaining a feature set through feature transformation, while feature selection consists in selecting a subset of features from the feature set. Both of these approaches can help to reduce feature dimensionality and redundancy. Feature extraction can sometimes reveal more meaningful feature attributes. In many cases, the feature selection process can indicate the importance of each feature for model construction.

Feature construction is the artificial construction of new features based on raw data. Feature extraction requires relevant domain knowledge to construct new, better and useful features. Feature extraction involves transforming features from the original data by some ready-made feature extraction methods, while feature construction requires manually constructing features, such as combining two features or breaking down a feature into multiple new features. Commonly used feature construction methods include [91]: the feature intersection, four fundamental operations of arithmetic, reconstruction of numerical quantities, decomposition of category features and decomposition of Data time.

If the feature engineering is done well, it will be easier or even unnecessary to adjust the later model's parameters, and the model's stability and interpretability will be better. Suppose the feature engineering is not done well, and the model evaluation cannot achieve the desired effect by adjusting parameters. In that case, it will take a lot of time to repeat processing and filter features until the model reaches the desired result. Therefore, feature engineering is a highly cumbersome and essential thing, which takes at least 80% of the modeling time. Consequently, it needs to be done patiently and logically to improve the performance and interpretability of the model.

2.3. Model selection and training

According to the different data types, machine learning algorithms can be divided into a supervised learning algorithm, unsupervised learning algorithm and reinforcement learning algorithm [92]. Supervised learning requires a clear label for each training dataset, and supervised learning is often used to deal with classification and regression problems. If the target characteristic is a continuous quantity (for example, superconducting transition temperature, resistivity), the process is called regression. Typical regression algorithms are support vector machine (SVM) [93], Gaussian process regression [94] or Kriging regression [95] and artificial neural network (ANN) [96]. If the output is a discrete target (for example, whether phase transition occurs or whether fracture occurs), the process is called classification. decision tree [97], random forest [98], naive Bayes [99] and K-Nearest Neighbor [100] algorithms are several commonly used classification algorithms.

Supervised learning aims to find functions that map a set of input data to corresponding output attributes, while unsupervised learning tries to identify the relationship between input data itself. For unsupervised learning, training data does not need to be labeled. In addition, unsupervised learning has no clear purpose training method, and it is impossible to know the result in advance, so unsupervised learning can hardly quantify the effect. Unsupervised learning is often used in clustering and dimensionality reduction. Clustering is a typical unsupervised learning method that divides the dataset into different categories or regions. The similarity between data points in the same group or cluster is higher than other clusters. In short, it is an automatic classification method [101]. In supervised learning, what each classification is exactly is known, but clustering is not. Clustering helps extract physical insights from data and discover new promising materials based on comparative studies [102]. The most popular clustering algorithms are hierarchical clustering [103], K-means clustering [104] and Markov random field [105]. The dimensionality reduction algorithm can solve the problems of sparse samples and difficult distance calculation of data points in high-dimensional space, which is convenient for feature extraction and

data visualization [106]. Different dimensionality reduction methods preserve different sample distribution characteristics in the low-dimensional subspace. The commonly used dimensionality reduction methods include multidimensional scaling (MDS), principal component analysis (PCA) [107], local linear embedding (LLE), linear discriminant analysis (LDA) and isometric mapping (Isomap) [108,109].

Finally, reinforcement learning is a rapidly emerging field. Unlike supervised learning, the input data of reinforcement learning is directly fed back to the model, and the model will adjust immediately. The classic application cases of reinforcement learning include nonlinear two-stage pendulum system (nonlinear control problem), chess games, robot learning to walk, unmanned driving, machine translation, and human-computer dialog [110,111]. The problem that can be solved by reinforcement learning is a sequential decision-making problem, which is a problem that requires continuous decision-making to achieve the ultimate goal. Compared with other machine learning methods, reinforcement learning focuses on goal-oriented learning from interaction. This learning model can explore the space of compounds and substances, such as finding new molecules by using objective-reinforced generative adversarial networks [112] and exploring grain boundary structure by using an evolutionary algorithm [113]. The commonly used reinforcement learning algorithms include online learning [114] and adversarial learning [115].

Fig. 3 summarizes the important ML methods. The above three paragraphs briefly introduce how machine learning algorithms are classified and the typical algorithms for each classification. Each algorithm's working principles and usage scenarios can be viewed in detail in these three classic books [116–118]. Over the past few years, many open-source software or tools have been developed, such as MatLab [119], Chainer [120], TensorFlow [121] and Scikit-learn [122], which enable non-professionals to implement ML methods. Many ML algorithms are also integrated into toolboxes in the commercial MATLAB software (such as statistics toolbox, machine learning toolbox and deep learning toolbox). As a powerful tool for neural network construction, Chainer aims to bridge the gap between algorithms and implementations. TensorFlow is a software library for high-performance numerical

computing that provides powerful ML and deep learning support. Scikit-learn is a Python package that integrates various unsupervised and supervised ML algorithms. Moreover, machine learning methods usually combine various intelligent optimization algorithms used to optimize model parameters, such as particle swarm optimization [123], genetic algorithm or simulated annealing algorithm. In addition, these optimization algorithms can also perform other difficult optimization tasks, such as the optimization of spatial configuration and material properties [124].

Furthermore, each algorithm has its characteristics, and there is currently no best algorithm for all problems. This problem is also known as the "free lunch theorem" [125]. The flow chart in Fig. 4 is the basis for the rough selection of the algorithm on the sklearn official website. The appropriate algorithm can be selected according to the following pictures and the actual situation of users. The pictures on the official website (https://scikit-learn.org/stable/tutorial/machine_learning_map/) have hyperlinks. Users can click to see the specific usage of many models [122]. Even experienced data scientists cannot judge which algorithm will be the best before trying different algorithms. We do not advocate a one-step approach, but we hope to provide some guidance on which algorithms to try according to some specific factors. (1) The size, quality, and type of data. Algorithms like neural networks can handle large amounts of data and features well. However, some algorithms, such as support vector machine, kriging and kernel ridge regression, can only handle a limited number of features [126]. Some models are less sensitive to outliers than others. Generally speaking, the tree model is not sensitive to the existence of outliers. However, regression models, or any model that attempts to use equations, are severely affected by outliers. In addition, some algorithms are designed to handle specific types of data. For example, the Naive Bayesian algorithm is particularly effective for processing the input to be classified, but it is completely insensitive to missing data. Images show a high degree of correlation between adjacent input points, and it may be the best choice to use a convolutional neural network to process image data [127]. (2) The interpretability of the algorithm. If the reason behind the result is the requirement of the problem, it is very important to understand the

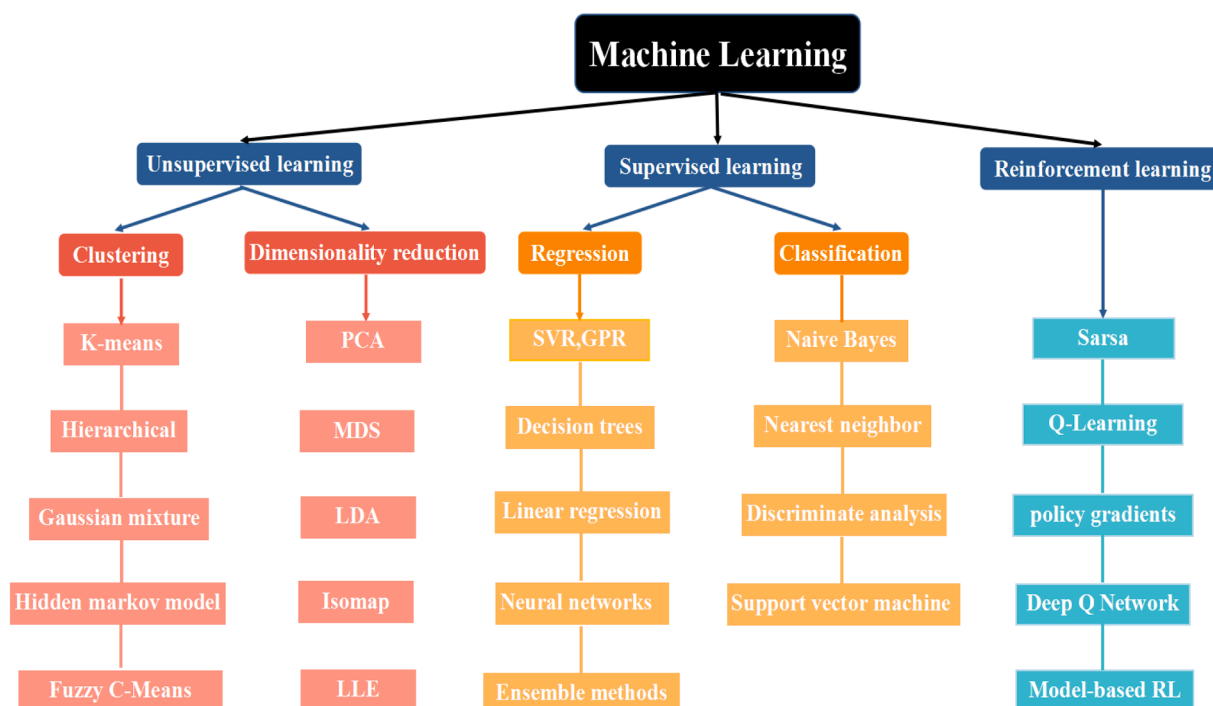


Fig. 3. Schematic representation of various ML algorithms. (PCA-principal component analysis, MDS-multidimensional scaling, LDA-latent dirichlet allocation, ISOMAP-isometric mapping, LLE-locally linear embedding, SVR-support vector regression, GPR-gaussian process regression).

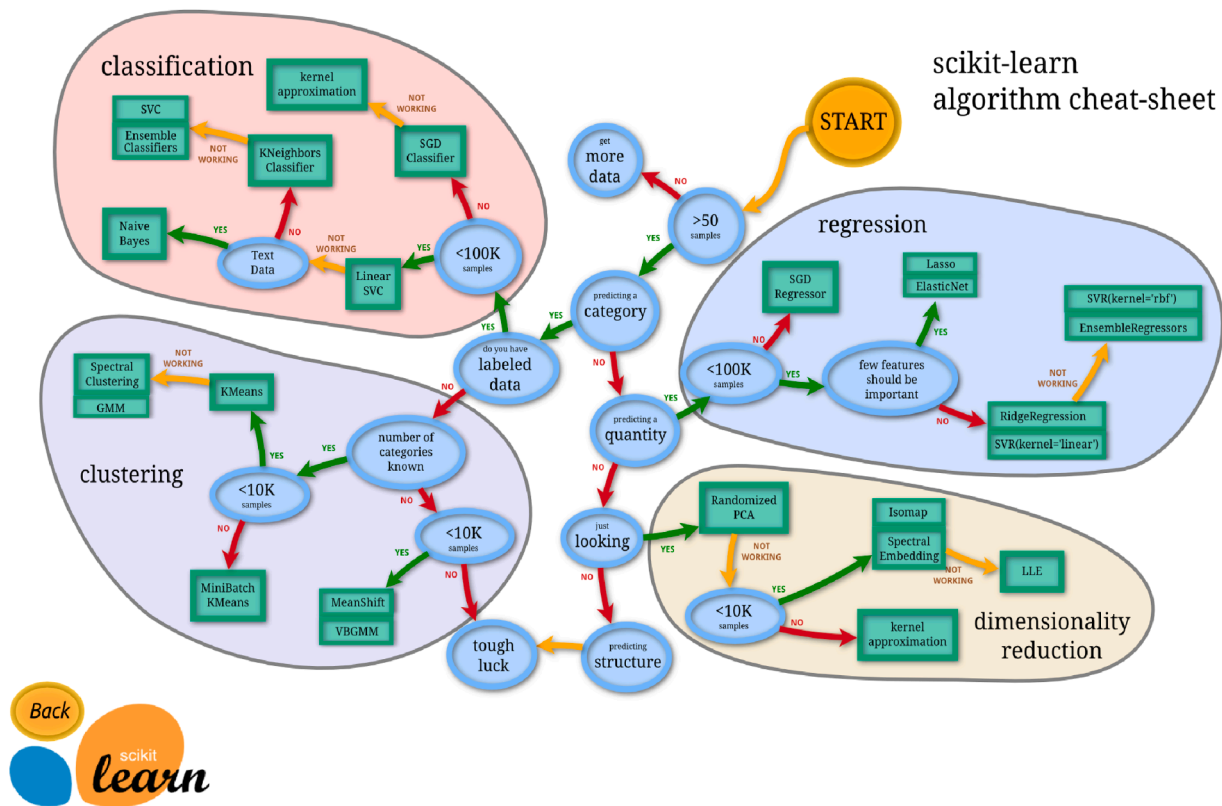


Fig. 4. Choosing the right estimator. Reproduced with permission [119] Copyright 2011 MIT Press.

interpretability of the algorithm. Algorithms such as k-nearest neighbor (KNN) are highly interpretable through feature importance, and algorithms such as linear models are interpretable through feature weight. However, the output process of algorithms such as neural network and gradient lifting algorithm is complex and abstract, which cannot be understood. (3) The urgency of the task. When the task time of machine learning is urgent, the training time and prediction time spent by the algorithm become extremely important. As we all know, neural networks need a lot of time to train a model. Traditional machine algorithms, such as k-nearest neighbor algorithm and logical regression algorithm, take much less time. Support vector machine, linear regression, logical regression and several types of neural networks can be used for fast prediction. However, algorithms such as KNN and ensemble models usually require more predicting time. Of course, in addition to the above three key factors, when selecting the most appropriate algorithm, we should also consider the types of tasks, the accuracy of the algorithm, the complexity of the algorithm, the data storage capacity, the scalability of the model and other factors.

2.4. Model evaluation and hyperparameter optimization

The choice of models exists in the field of machine learning, both in industry and in academia. How to choose a good model requires a set of criteria to evaluate the model so that we can choose. The evaluation methods and evaluation indicators of the model are mainly introduced here, which will be very useful. When there is only one dataset D and contains M samples, we can divide D into a training dataset S and a test dataset T , and the training set S is precisely divided into a training set and a validation set. Machine learning includes the training set and the verification set to build the optimal model. The optimal model evaluates the test error of the test set and evaluates the generalization ability of the learning method through the test error. Generalization ability is the gold standard to evaluate the performance of a machine learning model [128]. That is, the stronger the generalization ability, the better the

corresponding learning model. This paper introduces three commonly used model evaluation methods: hold-out method, cross-validation method and bootstrapping method.

In the "hold-out" method [129,130], the dataset D is divided into two mutually exclusive sets, one of which is training dataset S , the other is test dataset T , $D = S \cup T$, $S \cap T = \emptyset$. The training set S is divided into the training set and the verification set. The training set constructs the model, the verification set optimizes the model's parameters, selects the optimal model, and the test set T tests the generalization ability of the optimal model. A category-preserving sampling method called "stratified sampling" is often used to maintain the data distribution's consistency to avoid the impact of other deviations introduced by the data division process [131]. Since the relative ratio of S and T cannot be determined, $2/3-4/5$ of the samples in D are usually designated as S , and the rest are designated as T . Another problem with the hold-out method is that the training set S and the test set T are randomly sampled from the dataset D . Therefore, there are prominent contingency factors. It is necessary to perform the hold-out method multiple times to calculate the error rate of each test and then average the test error of each test to reduce the contingency factor. For example, Huang et al. [132] used machine learning to handle 28 feature descriptors and five properties of 153 high-energy density materials (HEDMs). Five machine learning models were trained and evaluated by stratified sampling and a 5-fold cross-validation method. Of these, the XGBoost model generated the best score for predicting HEDM properties.

"Cross validation" first divides the dataset D into k mutually exclusive subsets with the same size [133,134]: $D = D_1 \cup D_2 \cup D_3 \dots \cup D_k$, $D_i \cap D_j = \emptyset (i \neq j)$. Each subset D_i is obtained through stratified sampling, and then the union of $k-1$ subsets is used as the training set S , and the remaining subsets are used as the test set T . In this way, k groups of training/testing sets are obtained, and then K times of training and testing are carried out. Finally, the mean value of the K test results is returned. Generally, the cross-validation method is called "k-fold cross-validation method" [135]. The most commonly used value of k is

10, which is called 10-fold cross-validation. Leave-one-out cross-validation (LOOCV) occurs when k is equal to the number of samples m in D [136]. Because the training dataset of the leave-one-out method is only one sample less than the original sample dataset D , the evaluation result of the leave-one-out method is more accurate. However, when the training data is large, the computational cost of training m models may be intolerable (for example, if the dataset contains 1 million samples, 1 million models need to be trained). If the optimization of model parameters is considered, the amount of calculation will increase exponentially. Cross validation method is widely used in material design, for example, Alizadeh et al. [137] used Debye and transition temperature as descriptors and used cross-validation techniques to predict the target electron-phonon coupling constants for 28 elements. The result of our predicted electron-phonon coupling constant is consistent with the available value, with an accuracy of 88%. Sabnis et al. [138] used texture features and machine learning technology to classify the microstructure images of the four metals obtained by a light-optical microscope. Different steps such as image preprocessing, texture feature extraction and classification were applied to the images. According to the experimental results, XGBoost classifier achieves the highest classification accuracy of 98% thanks to the normalized texture features and 10-folds cross-validation method.

The "bootstrapping method" is sampling with putting back [139, 140], which randomly selects a sample from D each time, copies the sample into D' , and then puts the sample back into the initial dataset D . It is still possible to be collected in the subsequent sampling. Repeat this process for m times, and we get a dataset D' containing m samples, which results from bootstrapping sampling. Some samples of the initial dataset D will appear multiple times in the dataset D' , and some samples will not appear in the dataset D' . Finally, D' is designated as the training dataset, and D/D' is used as the test dataset. Because the number of training samples is equal to the size of the original dataset, the bootstrapping method is effective when the amount of data is small, and it isn't easy to correctly divide the training/test data. In addition, the bootstrapping method can generate multiple different training sets from the initial dataset, which has excellent benefits for methods such as ensemble learning (robust learning classifier). However, the bootstrapping method generates a dataset that changes the distribution of the initial dataset, which will lead to estimation bias. For instance, Chang et al. [141] used artificial neural network to predict the composition of high entropy alloy (HEA) based on non-equimolar AlCoCrFeMnNi to obtain the highest hardness in the system. Simulated annealing algorithms are combined with artificial neural network to optimize the composition. Bootstrap approach is adopted to quantify the uncertainty of prediction.

This paper discusses three evaluation methods of the machine learning model. Although the method of hold-out and cross-validation doesn't change the proportion of positive and negative analogies in the initial dataset through stratified sampling, the number of samples in the training dataset is less than that of the original dataset, and the probability distribution of the training dataset is different from that of the original dataset. Therefore, the estimation bias exists in the construction of the model by the two methods. Although the total sample size of the bootstrapping method is the same as that of the initial dataset, it changes the distribution of the initial dataset and the proportion of positive and negative categories. It also has estimation bias to replace the distribution of the initial dataset with the distribution of the bootstrapping sampling dataset. Therefore, it is recommended for the small sample dataset to use bootstrapping sampling method and then use a robust training classifier to construct the model. For a larger sample dataset, the cross-validation method is recommended, and for extra-large sample data, the hold-out method is recommended to construct the model.

When evaluating the algorithm, the appropriate evaluation indicators should be selected according to the types of the analysis problem. Correlation coefficient (R^2), Mean absolute percentage error

(MAPE) and Root mean square error (RMSE) are all used to evaluate the model to solve the regression problem [142]. The evaluation indexes for the accuracy of the classification model include Classification accuracy (CA), Recall rate, Accuracy, Receiver operating characteristic curve (ROC), Coen kappa value, Hinge loss, Logistic regression loss, Jaccard similarity coefficient, Confusion matrix, Coverage error, Hamming distance, Label ranking average accuracy and Ranking loss [143,144]. Rand index, Contour coefficient and Mutual information (MI) are indexes used for model evaluation in clustering problems [145,146].

The hyper parameter optimization is also called hyper parameter adjustment. Machine learning algorithms often contain millions of parameters, mainly including model parameters and hyper parameter. The hyperparameters are different from the general model parameters. The model parameters are the parameters learned by the model according to the distribution of training data, which do not need prior experience. However, hyperparameters are the parameters that are set before the learning process rather than the parameter data obtained through training. Machine learning algorithms such as SVM have hyperparameters such as gamma, kernel and ceof to adjust, while neural network models have more hyperparameters such as learning rate, optimizer and L1/L2 normalization [147]. There are two main difficulties in hyperparameter optimization: (1) Hyperparameter optimization is a combinatorial optimization problem and cannot be optimized by gradient descent methods like general parameters. (2) The time cost of evaluating a set of hyperparameter configurations (Configuration) is very high, which makes some optimization methods (deep neural networks) difficult to apply in hyperparameter optimization. A set of hyperparameters means that a deep neural network needs to be trained once, which will greatly increase the cost. See 4.3 The Complexity of Machine Learning for details on how to optimize hyperparameters.

3. Application of machine learning

Machine learning algorithms have different types and applications. The challenge is to determine which machine learning algorithms are closest to applications and to maximize prediction accuracy. In materials science, the application domains of machine learning algorithms are generally identified as predicting material properties, classification and identification, auxiliary micro-scale characterization, phase transformation research and phase diagram construction, process optimization, service behavior evaluation, accelerating the development of computational simulation technology, multi-objective optimization and inverse design of materials. The following is an introduction one by one.

3.1. Property prediction

The application of machine learning in material property prediction can be divided into two categories: macro-property prediction and micro-property prediction. The research on the macroscopic properties of materials mainly focuses on the mechanical, physical, magnetic and thermal properties of materials. The application of machine learning in microscopic property prediction tends to focus on several aspects, including lattice constants, structural energies, elastic moduli, phase volume fractions, band energies and electron affinity. According to the literature reports, the most widely used data information mining theory and technology is the modeling and prediction of the relationship between material composition, structure or process and properties. In this review article, we choose to focus on amorphous alloys, high-entropy alloys and superalloy materials to demonstrate the application of machine learning for macro- and micro-property prediction, respectively.

Amorphous alloy is a new type of metal material, also known as metallic glass or liquid metal, which is obtained by inhibiting the crystallization of atoms in alloy melts and maintaining and regulating the disordered structure of melt through new modern metallurgical technology in recent decades [148]. The research strategy of machine learning provides new ideas and methods for solving the core problems

faced in the field of amorphous alloys, which is conducive to promote the rapid development of the field of amorphous alloys. At present, the application of machine learning in the field of amorphous alloy property prediction mainly includes the following three aspects: (1) prediction of amorphous forming ability; (2) Prediction of mechanical properties of amorphous alloys; (3) Prediction of physical properties such as characteristic temperature and soft magnetic properties of amorphous alloys.

In terms of macro properties of amorphous alloys, Wang et al. [149] tried to predict the soft magnetic properties (saturation magnetization, magnetostriction coefficient, coercivity, Curie temperature, grain size and permeability) of Finemet iron-based nanocrystals by combining machine learning model (linear regression, support vector machine, decision tree, k-nearest neighbor and random forest) and global optimization. Taking crystalline silicon and amorphous silicon as examples, Qian et al. [150] developed a potential field based on machine learning to predict thermal conductivity. The interatomic potential based on machine learning is calculated by randomly sampling the potential energy surface in the configuration space according to the density functional theory, and then the thermal conductivity is calculated by molecular equilibrium dynamics. The results are in good agreement with the experimental results. In the future, with the development of high-throughput experiments, machine learning of amorphous alloys will achieve more remarkable breakthroughs. Keong et al. [151] designed a computational model based on back-propagation multilayer feedforward artificial neural network, the input parameters of the neural network model are alloy composition, heating rate and processing parameters, and the output parameter is the crystallization temperature (starting temperature, peak temperature and end temperature) of the main crystallization reaction of amorphous alloy. In the training and testing, the neural network has achieved good results in the prediction of crystallization peak temperature.

In terms of micro properties of amorphous alloys, Ren et al. [152] derived a criterion to characterize the correlation between glass forming ability and characteristic temperature based on the neighborhood principal component analysis method of feature selection. Finally, based on the classical nucleation theory, the feasibility of machine learning algorithms in glass forming ability research is proved. Logan et al. [153] developed a framework for designing metallic glass using a machine learning model. The model only takes the composition of alloy as input and predicts three key characteristics of the candidate bulk metallic glass: the ability of amorphous existence, critical casting diameter and supercooled liquid phase range. Xiong et al. [154] developed a prediction method using machine learning to describe the glass-forming ability and elastic modulus of bulk metallic glasses (BMG) based on a dataset of 6471 alloys. Peng et al. [155] proposed the random forest regression model for glass-forming ability (GFA) prediction based on the 810 data points in the alloy compositional dataset. The first step is to filter out various types of feature parameters related to GFA to determine the optimum features. Next, grid search is used to optimize hyperparameters for machine learning. Some studies have shown that the random forest regression model has great potential in predicting the glass formation energy of BMG.

High entropy alloys [156] are a new alloy design concept introduced in the early 21st century, in which elements are mixed in equiatomic or near-equiatomic ratios in an attempt to maximize configurational entropy. This structure gives high-entropy alloys excellent properties, including low-temperature toughness, strength and thermal stability at high temperatures, good corrosion and wear resistance, and good service performance under extreme conditions. For multi-component high entropy alloys, due to the large number of potential composition combinations, the traditional experience and trial-and-error method makes the development process slow and expensive, while machine learning accelerates the development of new materials by accurately predicting the properties of high entropy alloys, thereby effectively deal with the challenge of composition design of multi-component alloys.

In terms of macro properties of high entropy alloys, Wen et al. [157]

proposed a material design strategy that combines machine learning surrogate models with experimental design algorithms. After only seven experiments, an Al-Co-Cr-Cu-Fe-Ni system high entropy alloy with hardness 10% higher than the best value in the original training dataset was fabricated. Bundela et al. [158] trained and tested eight different machine learning algorithms to predict the microhardness of high entropy alloys. The dimensionality of high-dimensional data is reduced by principal component analysis to improve the prediction accuracy. The current study shows that the test R_2 score of XGBoost, Random forest and Bagging regression algorithm exceeds 0.89. Experimental data confirms the applicability of various training algorithms for performance prediction, and the ANN shows better performance on new experimental data. Based on 238 high entropy alloy data samples and 25 characteristic parameters such as alloy composition, modulus, density, mixing entropy and test temperature, Bhandari et al. [159] established a random forest model to predict the yield strength of high entropy alloy at different temperatures, with an accuracy of more than 95%. Xiong et al. [160] established a random forest model based on 71 tensile strength data samples and 290 hardness data samples. The predicted correlation between strength and hardness of high entropy alloy reached 0.95 and 0.9 respectively. Using Shapley weighted interpretation method, they analyzed the influence of different characteristics on alloy strength and found that there were critical values for three typical characteristics: valence electron concentration, mixing enthalpy and electronegativity difference.

In terms of micro properties of high entropy alloys, Chanda et al. [161] trained the neural network model by using 140 high entropy alloy data samples and taking seven characteristic parameters such as electronegativity difference, mixing enthalpy and mixing entropy as inputs, and the prediction accuracy of alloy elastic modulus is as high as 94%. As shown in Fig. 5, Kim et al. [162] explored the use of machine learning models to predict the elastic modulus of high entropy alloys. The ML model trained on a large dataset of inorganic structure can accurately predict the elastic properties of HEA. Roy et al. [163] implemented machine learning tools to predict the crystallographic phase and Young's modulus of low, medium and high entropy alloys composed of five refractory elements. The results and experimental verification show that the mean melting point and electronegativity difference play the most crucial role in the phase formation of these alloys, and the melting temperature and mixing enthalpy are the key features that affect Young's modulus. However, the effect of mixing entropy on the phase or Young's modulus is negligible. Dai et al. [164] used deep learning potential theory (DLP) to predict the mechanical and thermal properties of high-entropy ($\text{Zr}_{0.2}\text{Hf}_{0.2}\text{Ti}_{0.2}\text{Nb}_{0.2}\text{Ta}_{0.2}$), and the prediction errors of its force and energy were 217 meV/Å and 9.4 meV/atom, respectively. For TMC single phase carbide (TM=Ti, Zr, Hf, Nb, and Ta), DLP can predict the elastic constants and lattice parameters accurately. Gao et al. [165] adopted an extended finite element framework with multistage crystal plasticity (CPFEM) combined with an improved artificial neural network (ANN) algorithm to predict the dynamic and quasi-static uniaxial tensile mechanical response, texture and twinning characteristic in NiCoCrFe high entropy alloy. This result proves that the improved idea (ANN model optimized by GA) performs better than the CPFEM model based on physical mechanisms. Zhang et al. [166] proposed a robust data-driven framework based on Bayesian approaches for accurate and efficient structural energy prediction of high entropy alloys. Considering the limited data computed by first principles calculations, regularized Bayesian regression provides accurate and stable predictions and effectively quantifies the uncertainty associated with the effective pair interaction (EPI) parameters.

In the process of alloy design and optimization of superalloys, it is very important to accurately predict the properties, especially for those superalloys containing expensive alloying elements such as palladium, rhodium and ruthenium.

In terms of macro properties of superalloys, Khatavkar et al. [167] used the microstructures to estimate the hardness of Co- and Ni-based

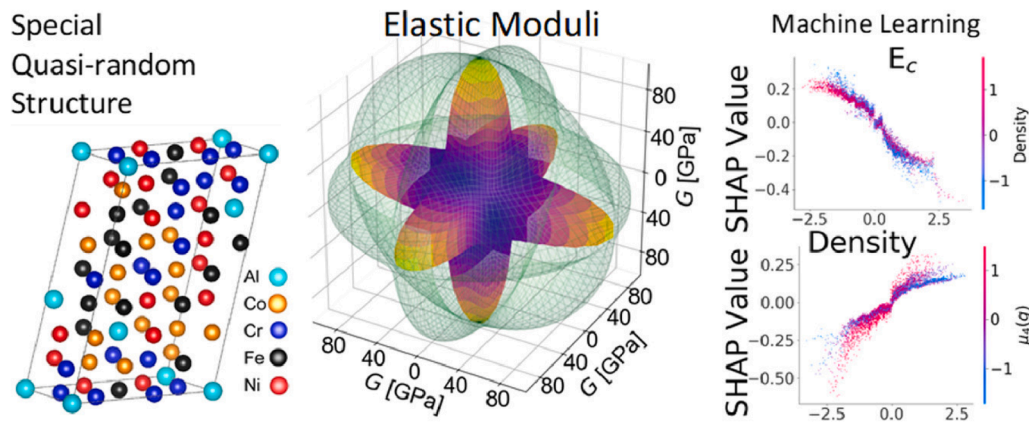


Fig. 5. First-principles and machine learning predictions of elasticity in severely lattice-distorted high-entropy alloys with experimental validation. Reproduced with permission [162] Copyright 2019 Elsevier.

superalloys. Complex image derived properties are utilized as a feature to develop a highly accurate ML model. Combining the characteristics of microstructure and composition, the relationship between vickers hardness, microstructure and composition parameters was established through Gaussian process regression (GPR), and the trained ML model showed unprecedented accuracy. Venkatesh et al. [168] used a back-propagation neural network to predict the creep life of INCONEL 690 superalloy at 1000 °C and 1100 °C with an accuracy of 100% and 90%, respectively. Liu et al. [169] embedded the micro characteristic parameters (such as stacking fault energy, lattice constant, phase volume fraction, etc.) calculated by calculation of phase diagram (CALPHAD) to build the creep life prediction model of Ni-based single crystal superalloy, and obtained a machine learning model with prediction accuracy higher than 91%. Wu et al. [170] used machine learning methods including linear regression, neural networks and multi-task elastic networks to predict the mechanical properties (ultimate tensile strength, micro-hardness and residual stress) of laser shock processing superalloy FGH4095. The results show that the predicted mechanical properties obtained by the machine learning methods are in good agreement with the experimental mechanical properties. As shown in Fig. 6, Ruan et al. [171] have successfully designed a new type of W-free Co-V-Ta-base alloys by employing CALPHAD method and a machine learning algorithm, which exhibit an extremely wide γ/γ' temperature range region, low mass density (8.67–8.86 g/cm³), a high intensity and a higher γ' solvus temperature (up to 1044 °C). Wang et al. [172] adopted the back-propagation artificial neural network model, and the

number of neurons in the input layer and output layer are 20 (18 kinds of alloy elements, test temperature and test time) and rupture strength respectively. The established model can accurately predict the rupture strength within a given alloy composition, test temperature and service life.

In terms of micro properties of superalloys, Jiang et al. [173] provided a machine learning approach to predict lattice misfit between the γ' and γ phase in Ni-based single crystal superalloys using relevant material descriptors including the measurement temperature, dendrite information and chemical composition and so on. The multilayer perceptron algorithm, sequential minimum optimization regression and support vector regression are performed on the experimental dataset to perform proper model selection, and the multilayered perceptron model performs well for its remarkable prediction performance. As shown in Fig. 7, Yu et al. [174] proposed an accelerated design strategy to find Co-base superalloys strengthened by γ' precipitates with good properties. The predicting models for four important properties (the existence of γ' and other phases, the γ' solvus temperature and area fraction) of Co-base superalloys were built first. Then, the four predicting models were integrated to design the Co-base superalloys that meet the designing requirements of four properties simultaneously. Zou et al. [175] developed an accelerated approach to design γ' -strengthened Co-Ni-base superalloys with well-balanced properties. First, a large number of experimental data was obtained using the diffusion-multiple method. Then, a machine learning model was established to construct the relationship between alloy composition and important

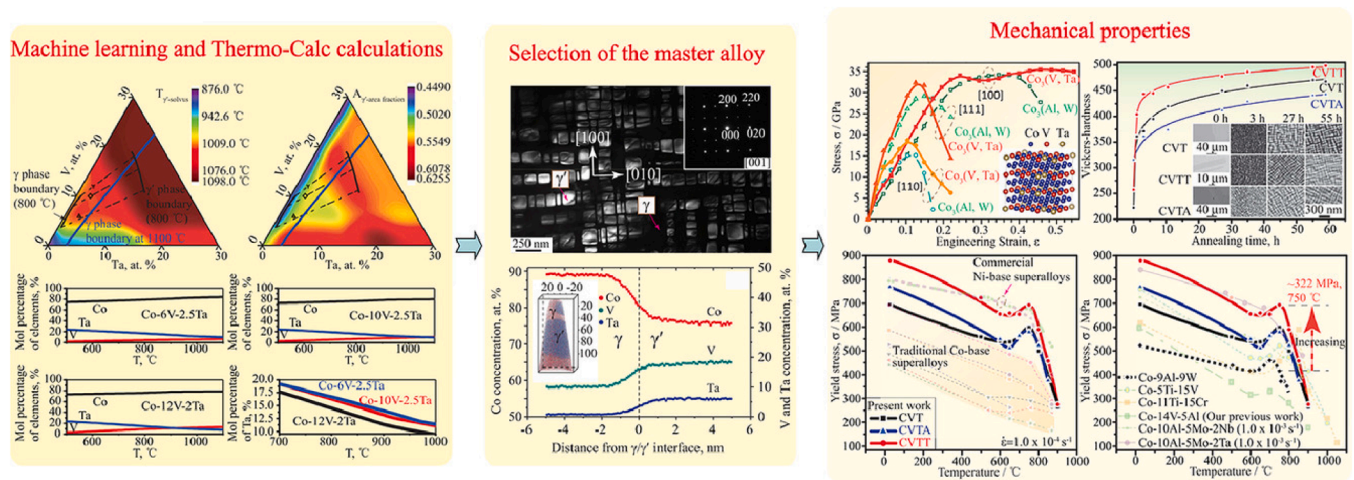


Fig. 6. Accelerated design of novel W-free high-strength Co-base superalloys with extremely wide γ/γ' region by machine learning and CALPHAD methods. Reproduced with permission [171] Copyright 2020 Elsevier.

Accelerated design of Co-base superalloys

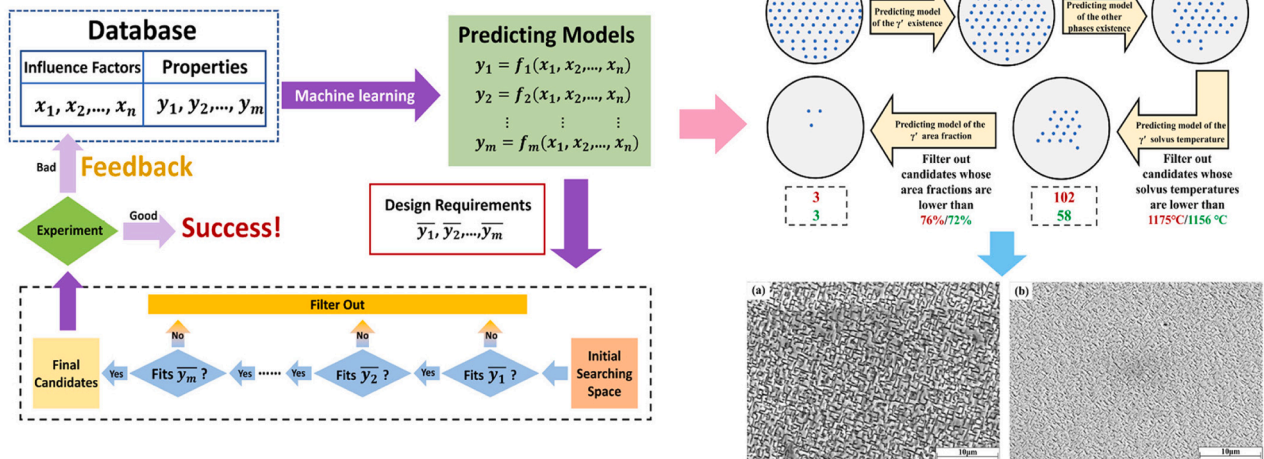


Fig. 7. Accelerated design of L12-strengthened Co-base superalloys based on machine learning of experimental data. Reproduced with permission [174] Copyright 2020 Elsevier.

thermodynamic and microstructure parameters, such as phase constituent, the γ' phase fraction and the γ' solvus temperature. Finally, based on the prediction results of the machine learning model of multicomponent Co–Ni-base superalloy and experimental verification, the composition optimization design of the alloy is realized. Zou et al. [175] obtained 1375 samples with material compositions of a γ/γ' two-phase region and the volume fraction of the γ' phase by using the Diffusion–multiple approach. Using the established database, a machine learning model that can accurately predict the volume fraction of γ' phase in superalloys is constructed through algorithms such as random forest and deep neural network. Tamura et al. [176] used machine learning to establish the relationship between gas atomization process parameters and grain size, and the obtained results were consistent with the trend in the formula proposed by Lubanska, so as to optimize the process parameters of Ni – Co based superalloy powder metallurgy process. Chandran et al. [177] used Bayesian ridge regression, ridge regression, support vector regression, LASSO, kernel ridge regression and linear regression to study the γ – γ' interface in Ni 617 superalloy, and predicted the energies of different configurations. All methods are superior to cluster expansion, and the error of kernel ridge regression is the smallest.

3.2. Classification and identification

The classification and identification of phase structure, microstructure, materials and defects do not depend on the important characteristics of domain knowledge. In recent years, significant progress has been made in this regard through the application of machine learning technology.

3.2.1. Phase structure classification

The classification algorithm of machine learning is widely used in the phase structure classification of superalloys [178–181], steel [182–184], high entropy alloys [185–188], etc. Taking a typical high entropy alloy as an example to illustrate how machine learning is applied to phase structure classification. High entropy alloys (HEAs) have attracted extensive attention due to their unusual properties, which largely depend on the choice of phases: amorphous, solid solutions, intermetallic compounds and their mixtures. Hence, accurate phase prediction is crucial to guide the selection of combinations of elements to form HEAs with desired properties. Huang et al. [189] used three different machine

learning algorithms: k-nearest neighbor, support vector machine and neural network algorithm to predict the phase structures of intermetallic compound (IM), solid solution (SS) and mixed IM and SS (IM + SS) of 401 high entropy alloys. Among which the neural network model has the best prediction effect, and the classification accuracy of solid solution is even as high as 94.3%. Nurat et al. [190] established the relationship model between alloy composition, characteristic parameters and phase composition by using artificial neural network. The model can quickly find and establish the quantitative relationship between alloy phase composition and intrinsic parameters such as electronegativity and atomic size difference, and judge whether single-phase solid solution, intermetallic compound or amorphous will be formed after different proportions of alloy elements are mixed. Zhang et al., [191] proposed feature selection and feature variable transformation based on kernel principal component analysis (KPCA), optimized feature variables, and used support vector machine (SVM) model for phase recognition. The results show that the F1-scores of the alloy phases considered are 0.9787, 0.9463, 0.9863 and 0.8103 respectively based on the four key feature variables and KPCA's SVM, which correspond to solid solution, amorphous, the mixture of solid solution and intermetallic, and intermetallic respectively. Based on the phase structure data of binary, ternary, quaternary and quinary equimolar alloys calculated by DFT, Kaufmann et al. [192] used the recursive feature elimination method to screen 108 composition features and 244 thermodynamic features to obtain 13 key features. Taking this as the input, they established a random forest model, and the classification accuracy of 134 experimental alloy phase structures (single-phase FCC, BCC or others) can reach 100%, which is better than the results of thermodynamic simulation and DFT calculation. Zhang et al. [55] proposed a systematic framework to effectively select ML models and material descriptors from a large number of alternatives using genetic algorithms (GA), and demonstrated its effectiveness in the Phase recognition of high entropy alloys (HEA). The classification model of radial basis function SVM allows the accuracy of identifying solid solution and non solid solution HEA reach 88.7%, and the accuracy of further identifying body centered cubic (BCC), face centered cubic (FCC) and two-phase HEA using neural network algorithm reaches 91.3%. In order to classify the high entropy alloy phase of solid solution, Machaka [193] trained and evaluated the eight classifiers such as regression tree, linear discriminant analysis, naive Bayes, generalized linear regression and random forest, and evaluated their classification. The result of feature selection determines

the most discriminating prediction feature. The accuracy of phase classification of RF, SVM, kNN and NNET classifiers is 97.5%, 95.8%, 94.5% and 94.0% respectively, which is superior to other algorithms.

3.2.2. Microstructure classification

Microstructure classification is mainly done manually by human experts, which will produce uncertainty due to subjectivity. Because the microstructure may be a combination of different phases or components with complex substructures, its automatic classification is very challenging. Image feature extraction and classification has been a basic and important research direction in the field of computer vision. Convolutional neural network (CNN) provides an end-to-end learning model. The parameters in the model can be trained through the traditional gradient descent method. The trained convolutional neural network can learn the features in the image, and complete the extraction and classification of image features. As an important research branch in the field of neural networks, convolutional neural networks are characterized by that the characteristics of each layer are obtained by the local region of the upper layer through the convolution kernel excitation of shared weights. This feature makes convolutional neural network more suitable for image feature learning and expression than other neural network methods. In recent years, convolutional neural network method has shown strong performance in the application of microstructure classification by combining classification steps to learn features from data. Taking some microstructure components of low carbon steel as an example, Azimi et al. [194] proposed a deep learning method for microstructural classification. The method employs a fully convolutional neural network (FCNN) and a maximum voting scheme for pixel-level segmentation. This system achieves a classification accuracy of 93.94%, which greatly exceeds the most advanced method of 48.89%. Based on the deep convolution neural network (DCNN) architecture, Pazdernik et al. [195] developed a fast and quantitative analysis method to classify various microstructure characteristics (grain, grain boundary, cavity, sediment and zirconia impurities) of unirradiated LiAlO₂ particles displayed by scanning electron microscope (SEM). Han et al. [196] proposed a high-throughput characterization method based on mathematical statistics, rapid acquisition technology and deep learning for the identification, segmentation and quantification of microstructure of weathering steel. The segmentation results show that the segmentation accuracy of pearlite phase and inclusion is 90.86% and 89.95% respectively. As shown in Fig. 8, Perera et al. [197] proposed an optimized machine learning framework, which uses convolutional neural network (CNN) classifiers to identify defects such as grain boundaries (GB), pores, or powder particles from a given microstructure. For GB,

RGB segmented images is generated using an optimized convolutional encoder-decoder network (CEDN). Another optimized CEDN is developed for the pore or powder particles to generate a binary segmentation. Microstructure classification methods based on deep learning have shown excellent performance in solving image classification problems, so they are widely used in microstructure classification. One limitation of applying deep learning models to microstructure classification, however, is that they require a tedious labeling process, especially for steel microstructures. Based on the rule-based segmentation method, Kim et al. [198] proposed a unsupervised inference algorithm captures the morphological features of each low-carbon steel microstructure and uses a Bayesian-Gaussian mixture model to automatically find the optimal number of microstructures with similar characteristics. This method can not only work without labeled images, but also does not need to know the number of microstructure components in each image in advance.

3.2.3. Material classification

Machine learning classification method greatly speeds up the search efficiency of new materials. Rahnama et al. [199] used a hydrogen storage material dataset to predict the optimal material class for metal hydrides based on the desired properties (pressure, operating temperature, heat of formation, and hydrogen-weight percentage). Four classification algorithms are employed: multiclass neural network, multiclass decision jungle, multiclass decision forest and multiclass logistic regression. It is shown that the multi-class neural network classifier has better classification performance achieving 0.80 accuracy. Based on a dataset of 397 ABO₃ compounds, Liu et al. [200] adopted machine learning to study stable and metastable perovskite materials. The best performance classification model Gradient boosting decision tree can classify 397 compounds into 143 non-perovskites and 254 perovskites with an accuracy of 94.6% over 10-fold cross-validation. Based on the hydride-metal hydrogen energy database publicly provided by the U.S. Department of Energy, Alireza et al. [201] employed unsupervised machine learning to identify the similarity of samples from different material categories. Isayev et al. [202] trained the data in AFLOW by using the gradient boosting decision tree of supervised learning after selecting the descriptor, finally realized the classification of metal and insulator, and successfully predicted the material band gap, bulk modulus and shear modulus, Debye temperature and heat capacity.

3.2.4. Defect identification

The problem of defect recognition is a research hotspot in the field of pattern recognition. An efficient and reliable defect recognition method

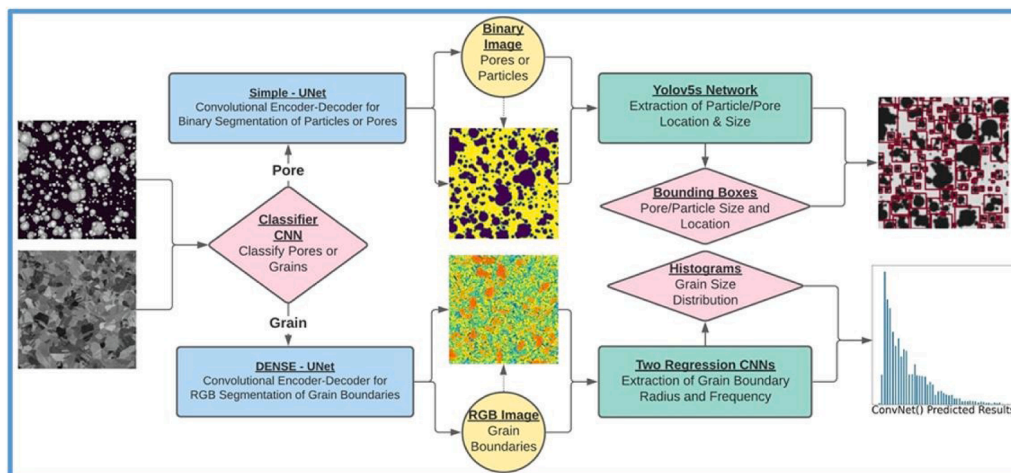


Fig. 8. Optimized and autonomous machine learning framework for characterizing pores, particles, grains and grain boundaries in microstructural images. Reproduced with permission [197] Copyright 2021 Elsevier.

can effectively improve efficiency and maintain safety. For common nondestructive testing methods, some are mainly used to detect surface defects and damages of workpieces, such as magnetic particle testing and penetrant testing; Some are used to detect internal defects and damages of workpieces, such as ultrasonic testing and X-ray testing. Based on the classification of inspection methods, the problem of defect identification can be roughly divided into surface defect identification and internal defect identification. Examples of machine learning for surface defect recognition include: Lan [203] first used the neural network method to identify the defects on the workpiece surface, based on its low accuracy, it was improved, and some parameters in the neural network were improved using particle swarm algorithm, and finally achieve the purpose of improving the accuracy of defect recognition. Cui et al. [204] proposed a method of laser ultrasonic metal workpiece surface defect recognition based on random forest, which effectively improved the accuracy of metal workpiece surface defect recognition by using random forest. Feng et al. [205] used deep neural network to identify solidification defects. By establishing deep neural network with multiple hidden layers and multiple input variables, and comparing the training results of deep neural network with traditional shallow neural network such as support vector machine and artificial neural network, it is proved that deep neural network is also suitable for dealing with material defects in small samples. Detecting surface defects is a challenging visual recognition problem arising in many processing steps during manufacturing. Weimer et al. [206] presented a machine vision system which uses basic patch statistics from raw image data combined with a two layer neural network to detect surface defects on arbitrary textured and weakly labeled image data. Evaluation on an artificial dataset with more than 6000 examples in addition to a real micro-cold forming process showed excellent classification results.

In terms of internal defect identification: Real-time weld defects detection is still challenging due to the complexity and diversity of weld defects. Zhang et al. [207] proposed an online detection method for robot arc welding defects in aluminum alloys based on arc spectroscopy and random forest. A defect identification model based on optimal feature subset and random forest is established. Compared with BP and RBF, the model can better identify three typical defects, including lack of porosity, burn-through, and penetrance. Boaretto et al. [208] proposed a method for automatic detection and classification of defects in radiographic images of welded joints using double wall double image exposure technology (DWDI). In this method, the feedforward multilayer perceptron is used to build a workpiece weld defect detection classifier, and the back propagation learning algorithm is integrated to realize the automatic detection of weld defects. Gaolong et al. [209] developed an automatic and reliable non-destructive evaluation (NDE) technology, which can simultaneously quantify the width and depth of defects under the surface of metal parts by using non-contact laser ultrasonic technology and recognition machine learning algorithm. Based on the analysis of light intensity data on site from the melt pool during laser powder bed fusion (LPBF), Katayoon et al. [210] used the self-organizing mapping (SOM) algorithm to detect the porosity due to the lack of fusion (LoF) phenomenon in the process of laser powder bed fusion. Compared with the traditional prediction algorithm (such as absolute limit), the proposed SOM algorithm has higher prediction rate in the case of faster speed, and can quickly identify the location of defects. Transmission electron microscopy (TEM) is a common method to characterize and quantify material defects. The analysis of digitized TEM images is usually done manually, which is a time-consuming and error prone task and cannot be extended to large data sets. Jacobs et al. [211] used the mask R-CNN model of deep learning to perform semantic segmentation for various defect types in the electron microscope image of irradiated FeCrAl alloy, thus promoting the development of automatic methods for defect quantification and analysis of TEM images.

3.3. Auxiliary micro-scale characterization

The microstructure of a material directly affects its macroscopic properties. The characterization and regulation of the microstructure is the key means to develop the theoretical basis of materials and the design and development of new materials. The traditional characterization ranges from the early metallographic structure to the analysis of crystal structure and primary defects with the help of scanning electron microscope and transmission electron microscope, as well as the latest in-situ analysis, environmental electron microscope and other more sophisticated observation methods. In addition to relying on advanced experimental equipment, these research methods also require researchers to have rich calibration experience, which brings great challenges to the characterization and analysis of micro scale. In recent years, the application of machine learning methods such as image recognition, classification algorithm and dimension reduction processing to the structural characterization and analysis of materials plays a great role in promoting the understanding of the theoretical basis of materials and the research and development of new materials [212]. At present, machine learning assisted micro-scale characterization analysis is mainly applied in the following three aspects: (1) image analysis using machine learning methods to identify structures; (2) search the relationship between microstructure and material properties; (3) improve microstructure characterization technology.

The recognition and analysis of microstructure is a very important part in the research of metal material structure. With the help of image recognition and classification technology of machine learning, the microstructure can be recognized and characterized quickly. Oktay et al. [213] proposed a new method to detect nano-particles and determine their shape and size simultaneously with deep learning. This method employs a multi-output convolutional neural network and has two outputs: one is the detection output that provides the particle's position, the other is the segmentation output that provides the boundary of the nano-particle. Yang et al. [214] developed a machine learning model to automatically identify 2D material sheets with different layers from optical microscope images and then determine their transverse dimensions. The statistical distribution of the sheet size is obtained and used to estimate the strength of the correlated 2D material according to the distribution-property relationship we developed previously. DeCost et al. [215] applied a deep convolutional neural network segmentation model to perform pixel-level segmentation on a dataset of ultra-high carbon steel microstructures. Two microstructure segmentation tasks are mainly explored: segmentation of cementite particles in a spheroidized matrix, and segmentation of microstructures containing features including grain boundary carbides, spheroidized particle matrix, particle-free grain boundary spalling zones, and Widmanrite. This provides a new avenue for automated microstructural segmentation applications of complex microstructures. Automatic identification of nanoscale crystallographic defects in electron micrographs controlled by complex contrast mechanisms remains a challenging task. Roberts et al. [216] introduced DefectSegNet, a new convolutional neural network architecture, which can segment three common crystal defects in structural alloys: dislocation lines, precipitates and voids. The supervised training results of a small group of high-quality steel defect images show that the pixel accuracy of all three defects is very high: the dislocation is $91.60 \pm 1.77\%$, the precipitate is $93.39 \pm 1.00\%$, and the gap is $98.85 \pm 0.56\%$. Madsen et al. [217] developed a deep learning-based algorithm to identify local structures in TEM images, which is stable to microscope parameters and noise. Neural networks can reliably predict experimental images through training. The method is applied to identify the defective single graphene and the metal nanoparticles on the oxide carrier, and the identification results of the microstructure demonstrated the effectiveness of the method. Ziatdinov et al. [218] showed a general framework based on deep learning for locating and characterizing atomic objects in a lattice. Specifically, by examining and classifying the features in the output layer of the convolutional neural network, it can

detect the structure related to the existence of point defects in the tungsten disulfide monolayer, and strongly correlate the inhomogeneity of the charge density distribution around the specific lattice position on the oxide surface. Alrfou et al. [219] applied a machine learning method to automatically measure the area fraction of graphite particles in the microstructure of ductile iron. In the unsupervised learning step, K-means clustering is used to generate the segmentation masks as the training data of the supervised learning step. For the supervised learning step, the U-Net structure (a fully convoluted neural network) is used to establish the model and predict the area fraction of graphite.

In order to fundamentally improve the properties of materials, we must explore the microstructure of materials, it is necessary to explore the microstructure of materials. Machine learning technology plays a great role in retrieving the relationship between microstructure and properties. As shown in Fig. 9, Kondo et al. [220] applied a convolutional neural network to link the experimental microstructure with the corresponding ionic conductivity. The results show that the convolutional neural network can be trained by using only seven micrographs, and its performance is better than that of the traditional scheme using hand-made features. Liu et al. [221] established the relationship between the microstructure and mechanical properties using an artificial neural network model based on the microstructure and mechanical properties data of rolled C-Mn steel. The microstructure includes volume fractions of ferrite, pearlite, and austenite, and grain size of ferrite. The predicted mechanical properties include elongation, yield strength and tensile strength. Khatavkar et al. [167] used image recognition technology to extract material microstructure information as descriptor. Firstly, the SEM images of two nickel-based superalloys, cmsx-6 and DZ125, are processed to produce images with only black and white colors. Then the image information is extracted by image recognition technology. The author applies this descriptor to machine learning and achieves the accurate prediction of nickel-based superalloy hardness. Suh et al. [222] investigated the quantitative correlation between the microstructure, texture and mechanical properties of aged AZ61 and AZ91 bars by machine learning. In order to simplify the generation of data, only SEM images are used to quantify the features of discontinuities and continuous precipitation. The artificial neural network is used to predict the tensile and compressive yield strength and the resulting yield asymmetry. A total of 288 data sets are predicted with 11 prediction factors, and the accuracy is more than 98%. For heat-resistant alloys, the prediction of creep properties is crucial but still very challenging, and data-driven materials genomics technology provides a promising approach for heat-resistant alloys. As shown in Fig. 10, Cecen et al. [223] developed a data-driven method to accurately associate the 3D microstructure with its homogenized properties. An extremely large

benchmark microstructure-property dataset of 5900 microstructures is created, and it is demonstrated that the CNN based approach cannot only learns interpretable microstructural features but also leads to improved accuracy in property predictions for new microstructures. Based on textual (chemical composition and creep test conditions) and visual (as cast microstructure) data, Xiang et al. [224] explored the application of deep learning to quantitatively predict the creep fracture time of Fe-Cr-Ni heat resistant alloys. The textual and visual data were analyzed using a multilayer perceptron (MLP) and a convolutional neural network (CNN), respectively. The multi-source heterogeneous data is then fused using a fusion deep learning model, which has a significant improvement in prediction accuracy compared to a single MLP and CNN model.

In addition, machine learning also offers several novel avenues in assisting microstructural characterization. The availability of high-quality 3D microstructures is a necessary prerequisite for understanding the macroscopic behavior of heterogeneous medium. Considering the high cost of 3D microscopic imaging, it is an economical method to obtain a large number of virtual 3D microstructures from limited morphological information of 2D cross-sectional images. Fu et al. [225] proposed an effective method to statistically reconstruct 3D microstructure by combining 2D image features based on machine learning. The results show that this method provides a low-cost and widely applicable method to reproduce the 3D microstructure, and accurately retains the statistical features, geometric irregularities, long-distance connectivity and anisotropy in the 2D cross-sectional image. As shown in Fig. 11, Shen et al. [226] developed two convolutional neural network methods based on region transformation and transfer learning to reconstruct crystal orientation from electron backscatter diffraction patterns (EBSD). The model has strong robustness to the image noise and index direction measured in the experiment and is as fast as the highest EBSD scanning rate. Ding et al. [227] proposed a new convolutional neural network (EBSD-CNN) to achieve a real-time index of EBSD patterns, which implements a disorientation loss function to adapt a standard CNN model for crystallographic orientation indexing. The results show that CNN can provide an index method that replaces the commercial Hough transform with higher accuracy and speed. Tran et al. [228] proposed a microstructure reconstruction method that can generate images with the fidelity of an experimental microscope without introducing any physics-based microstructure descriptors. Furthermore, the image texture is preserved, and continuous variables are used instead of binary or categorical variables to represent the microstructure image, which results in a high-fidelity image of microstructure reconstruction. Background subtraction remains a great challenge especially in the environment where multiple sources of background signals

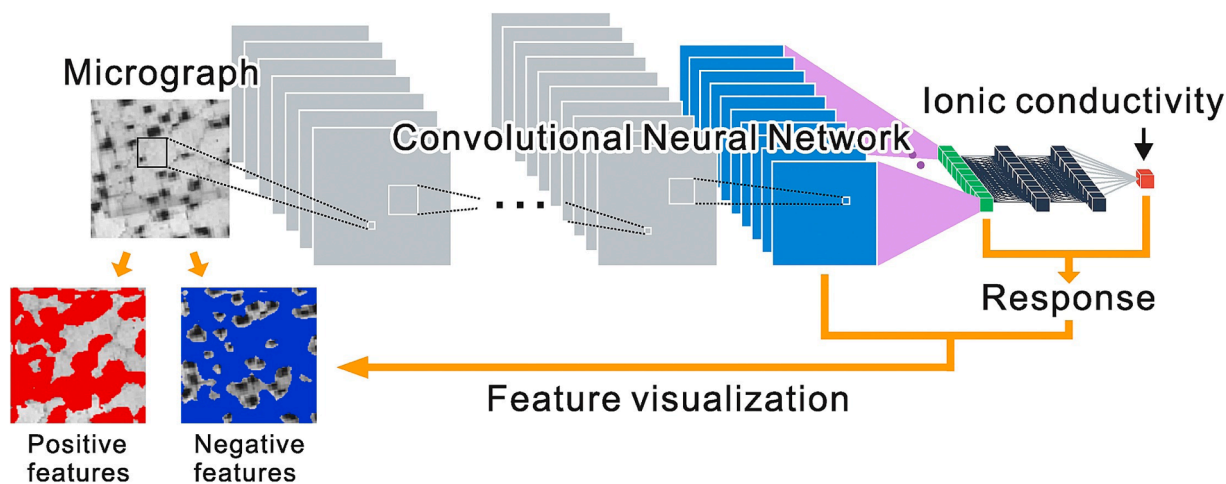


Fig. 9. Microstructure recognition using convolutional neural networks for prediction of ionic conductivity in ceramics. Reproduced with permission [220] Copyright 2017 Elsevier.

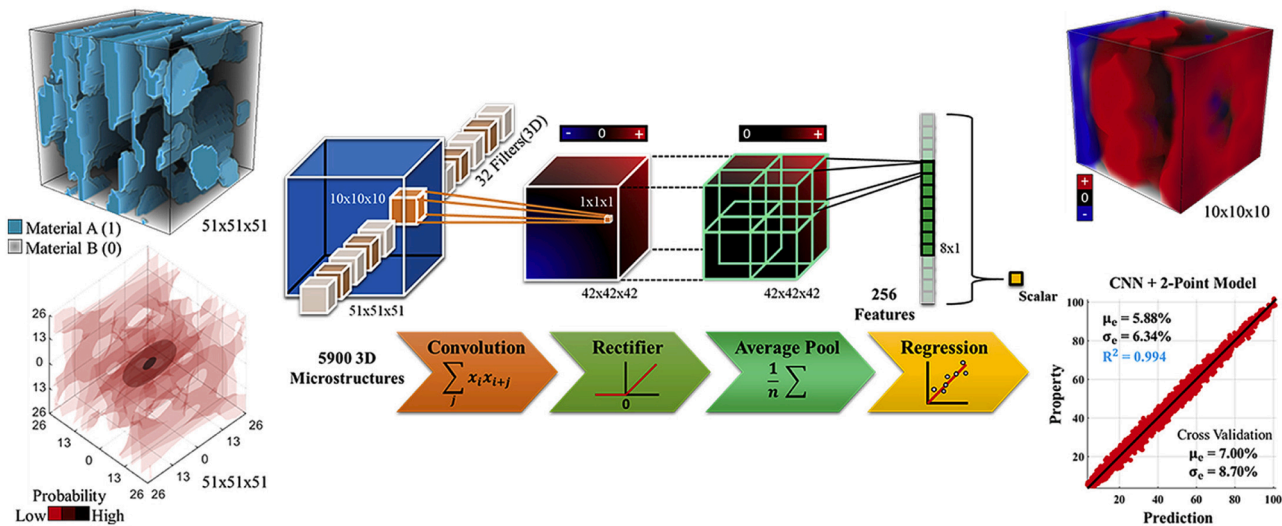


Fig. 10. Material structure-property linkages using three-dimensional convolutional neural networks. Reproduced with permission [223] Copyright 2018 Elsevier.

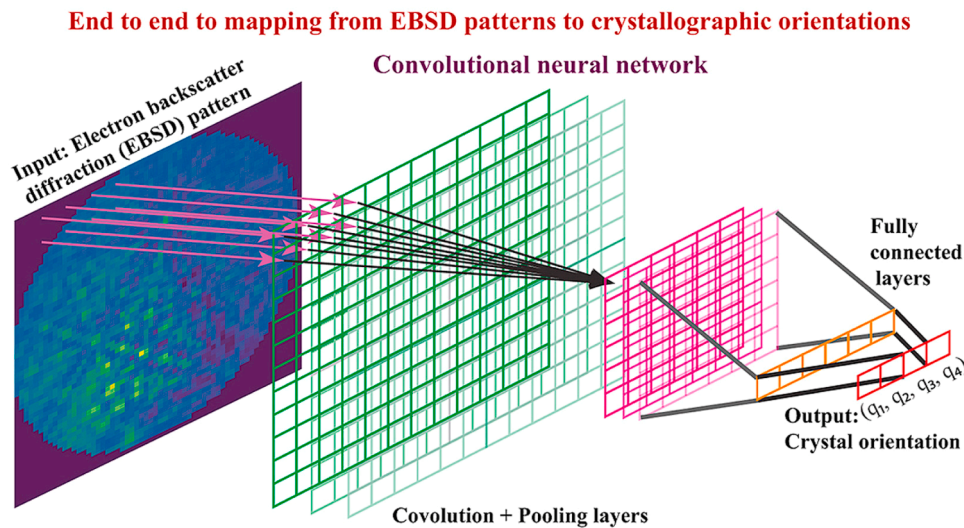


Fig. 11. Convolutional neural network-based method for real-time orientation indexing of measured electron backscatter diffraction patterns. Reproduced with permission [226] Copyright 2019 Elsevier.

coexist. Here, Ament et al. [229] propose an unsupervised probabilistic learning method, which scans large datasets to identify multiple background sources and determine the likelihood that any given data point contains a signal of interest. The method has been demonstrated on Raman spectroscopy and X-ray diffraction data. Zhu et al. [230] proposed an inspection pipeline that integrates (i) miniature X-ray computed tomography reconstruction, (ii) automatic three-dimensional morphological analysis, and (iii) machine learning-based big data analysis. Based on these, machine learning analysis is employed to reveal quantitative insights into the global evolution of defect features beyond the qualitative observation of humans.

3.4. Phase transition research and phase diagram construction

The application of data mining technology in phase transformation research mainly includes the prediction of the isothermal transformation (Time-Temperature-Transformation, TTT) curve and continuous transformation (Continuous-Cooling-Transformation, CCT) curve of metal materials, Ac1 and Ac3 temperature, martensite Transformation starting point Ms, austenite starting temperature As,

hardening curve, etc. In addition, machine learning helps to build phase diagrams quickly and efficiently.

Time temperature transition (TTT) diagram and continuous cooling transition (CCT) diagram are important basis for formulating the optimal heat treatment process of steel with ideal microstructure and properties. However, the experimental determination is time-consuming and laborious, and the calculation method is limited by the composition [231,232]. Therefore, it is of great practical significance to predict TTT and CCT diagrams quickly and accurately, especially for expensive and time-consuming material design. Huang et al. [233] developed a combined machine learning (CML) model to predict the TTT diagram of high alloy steel. Firstly, a classification model is established, and the shape of the TTT graph is determined by the filtered classifier. Then, the regression model is developed by combining random tree, random forest, bagging, k-nearest neighbor and random committee to quantitatively predict the TTT diagram of high alloy steel. The results show that the CML model has higher correlation coefficients and prediction accuracy with lower error values compared to JMatPro and empirical formulas, which shows the advantages of machine learning method in TTT chart prediction. Meanwhile, as shown in Fig. 12, Huang et al.

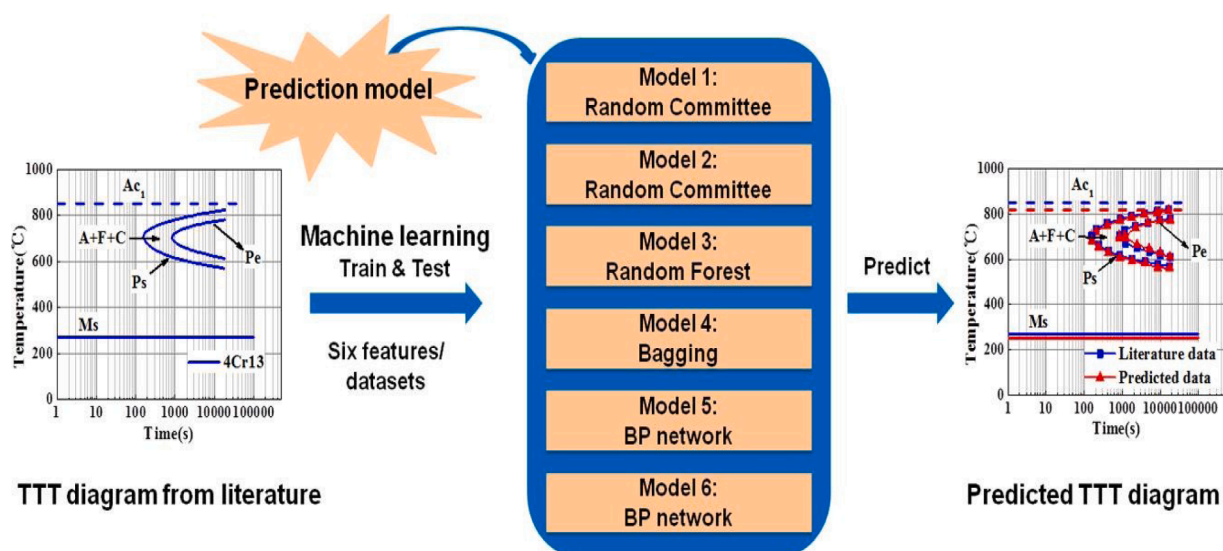


Fig. 12. Study on time-temperature-transformation diagrams of stainless steel using machine-learning approach. Reproduced with permission [234] Copyright 2020 Elsevier.

[234] also developed a combination of machine learning (ML) algorithms including bagging, random forest, random committee and BP artificial neural network to predict TTT diagram with associated descriptors containing alloying elements, holding time and austenitizing temperature. The results show that this combination can achieve high prediction accuracies on the TTT diagram of stainless steel. Malinov et al. [235] designed an artificial neural network model for the simulation of time-temperature-transformation (TTT) diagram of titanium alloy. Data from published literature are used to train standard back-propagation multilayer feedforward networks. Based on the trained neural network, the effects of molybdenum, vanadium, oxygen, and aluminum on the phase transformation kinetics of titanium alloy are investigated, and the obtained results are consistent with the expected results of phase transformation theory. In order to predict the CCT diagram of tool steel, Geng et al. [236] used relevant material descriptors (cooling rate, chemical composition and austenitizing temperature) and different machine learning techniques (random forest, bagging, k-nearest neighbor and multilayer perceptron regressor, etc.). The k-nearest neighbor and the bagging methods are suitable for predicting the starting and ending temperatures of bainite formation, respectively. The random forest has been proven to be the best model for accurately predicting pearlite transformation temperature and martensite transformation onset temperature (M_s). In order to predict the hardness and transformation temperature of low alloy steel with cooling rate and chemical composition, Geng et al. [237] established a hybrid machine learning model consisting of random forest, k-nearest neighbor, and multilayer perceptron classifier. In this paper, the SH-CCT diagrams of six kinds of steel are computed using hybrid machine learning model. The results show that the classification model is 100% accurate.

The temperature of Ac_1 , Ac_3 , M_s and A_s are the important phase change points required when formulating the heat treatment process [238]. Based on the percentage of alloying elements in the chemical composition of steel, Masoud et al. [239] respectively used the neural networks of feedforward topology and Bayesian regularization training algorithm to predict the austenite formation temperature (Ac_1 and Ac_3). The results show that the Bayesian regularization neural network has the best performance. You et al. [240] developed an artificial neural network model for predicting the austenite formation temperatures (Ac_3 and Ac_1) of steels. Scatter plots and statistical criteria show that the prediction performance of the artificial neural network is better than that of the Andrews formula. Wang et al. [241] predicted M_s temperature of high carbon steel by artificial neural network. At the same time, it

enables us to estimate the quantitative effect of alloying elements on M_s temperature on a considerable optional scale. The combination of multilayer feed-forward neural networks (MLFFNNs) with knowledge of thermodynamics has excellent advantages for studying complex phenomena in materials science. Lu et al. [242] developed an MLFFNN based on thermodynamic knowledge to predict the martensitic onset temperature (M_s) of steel, which explains the changes in the critical driving force (ΔG_c) and the austenitization temperature (T_γ). Adjusting the martensitic transformation temperature through composition design has become an important way to broaden the applicable temperature range of shape memory alloys (SMA). As shown in Fig. 13, Xiong et al. [243] proposed an empirical formula enhancement strategy based on a machine learning approach to establish the relationship between the composition and the martensitic transformation onset temperature (M_s) in Cu-Al based SMA systems. Based on the rich augmented dataset combined with the predicted data of the SVR.rbf model and the experimental data, the enhanced empirical formula with M_s as a component function was fitted. A back propagation artificial neural network (ANN) model of a gradient descent learning algorithm was built, trained and tested by Eyercioglu et al. [244] using test data of 85 iron-based shape memory (SMAs) alloys available from the literature. For the artificial neural network model, the input parameters are the weight percentages of seven elements and three different processing conditions (quench, homogenization temperature and warm rolling). They found that the ANN model predicts the temperatures M_s and A_s well within the considered input parameter range.

Dong et al. [245] conducted a comparative study on the establishment of chemical composition-hardness models of wear-resistant steel using six machine learning methods such as artificial neural network, gradient boosting regression, and random forest. The results show that among the six machine learning methods we studied, the artificial neural network method with a $32 \times 32 \times 32$ structure has the highest prediction accuracy. Prediction of gear steel hardenability and chemical composition was studied using artificial neural networks. Gao et al. [246] used software to quantitatively predict the hardenability of its chemical composition or the chemical composition of its hardenability. The forecast results are more accurate than those obtained by traditional methods based on simple mathematical regression models.

Phase diagrams are essential tool for optimizing heat treatment processes, but the experimental determination of phase diagrams is time-consuming and expensive, and the computational methods are also limited by the composition [247]. Therefore, material researchers have

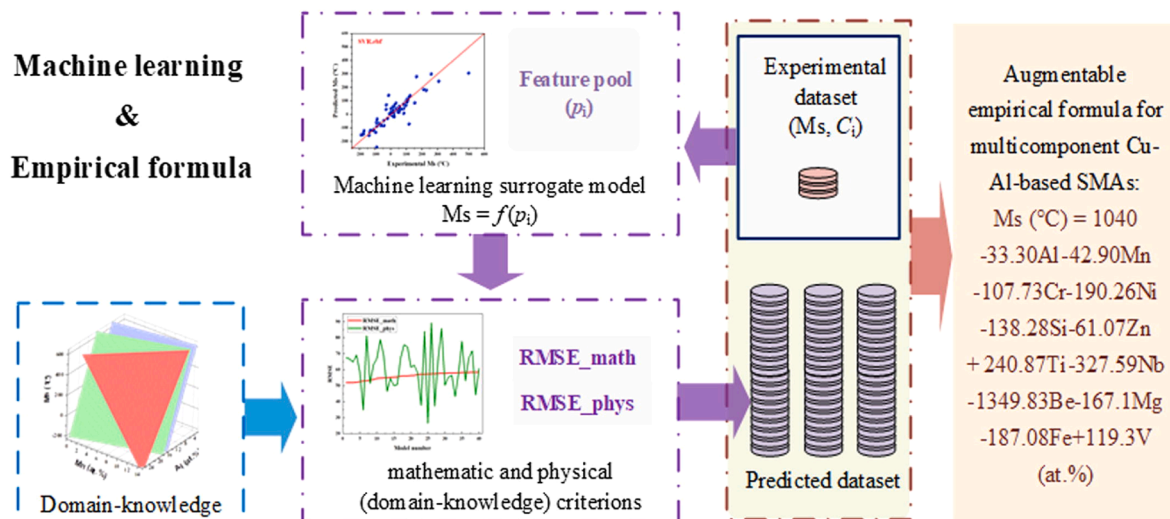


Fig. 13. Machine learning assisted empirical formula augmentation. Reproduced with permission [243] Copyright 2020 Elsevier.

turned to the rapid data analysis technology of machine learning to convert a large number of structured data into phase diagrams [248–252]. In particular, the active learning strategy in machine learning has achieved success in the construction of phase diagram. As one of the most popular learning strategies in machine learning, active learning has been successfully applied to new material discovery [253–255], efficient computation of materials [256,257], guiding experimental design [258,259], phase diagram construction and other fields. In terms of phase diagram construction, using active learning strategy to supplement new data samples with higher yield can quickly improve the accuracy of machine learning classification model and effectively reduce the amount of data required to draw phase diagram. To efficiently construct the phase diagram of the alloy system, Terayama et al. [260] proposed a thermodynamic phase equilibrium method based on machine learning method. Using uncertain sampling in active learning can recommend the next point to synthesize or measure for effective phase mapping. Compared to random sampling, the recommendations of this machine learning method help reduce the number of experiments needed to construct phase maps to about. Terayama et al. [261] also developed a method to efficiently construct phase diagrams using machine learning. Dense sampling is carried out around the phase boundary by an uncertain sampling strategy in active learning. Compared with random sampling, the method of uncertainty sampling reduces the number of sampling points to about 20%. In particular, the reduction rate is evident in more complex phase diagrams. Furthermore, this method requires only a small number of initial sampling points to quickly find undetected new phases. Although advanced sampling techniques such as Gibbs ensemble and Gibbs Duhem integral can effectively sample phase balance, they may not be extended to many unbalanced systems. Grid search suffers from inefficiencies by sampling predetermined state points that provide little information about phase boundaries. Dai et al. [262] proposed an active learning framework to overcome this defect and drastically reduce the sample size required to establish phase boundaries by adaptively selecting the next most informative state point each round. Based on the small-scale experimental data, Tian et al. [263] used the active learning strategy to predict and experimentally verify the phase diagram of multicomponent system from the high dimensional virtual space of all possible phase diagrams involving multiple elements. In this method, the machine learning method is used to predict the phase diagram and the Bayesian experimental design is used for minimizing experiments for optimization and verification.

3.5. Process optimization

The formulation of a reasonable material processing technology has always been based on theoretical analysis and experience accumulation. However, the established process parameters can not guarantee the purpose of improving the quality of the processing process and the properties of the workpiece. Therefore, it is necessary to optimize the design of the forming process. For those processes that are difficult to establish a model from the mechanism, the data mining technology is used to establish the relationship between the process parameters and the quality of the process or material properties. It can not only be used to optimize the process, but also be used to analyze the impact of process parameters on material properties, which provides a convenient method for the establishment of process model.

Bharti et al. [264] first explored the mixed application of artificial neural network and multi-objective optimization algorithm (NSGA-II) in cylindrical grinding of superalloys. The multi-layer perceptron neural network is associated with NSGA-II in the form of an objective function. Two grinding wheels are used, one is a conventional grinding wheel, and the other is a CBN super-hard abrasive, which optimizes the grinding process of superalloys. Singh et al. [265] processed Nimonic 75 (Ni-based superalloy) using electrical discharge machining. Teaching based learning optimization algorithm (TLBO) is used for multi-objective optimization of the performance indicators (material removal rate and surface roughness). The verification test is carried out to verify the results obtained by the TLBO method. An effective optimization method based on a support vector machine was proposed by Nametala et al. [266] to optimize multiple process parameters of selective electron beam melting (SEBM) of the non-weldable nickel-based superalloy ELEG. Extract optimal global conditions to manufacture SEBM samples. All SVM optimized conditions can produce near-net-shape samples with a uniform top surface. Based on the production data of the manufacturing plant that produces turbine blades in investment casting, Carmen et al. [267] adopted support vector machine learning methods to optimize the process of superalloy investment casting. In addition, some discussions on 3D printing, additive manufacturing and other material processing processes can refer to [268–273].

Zhang et al. [274] selected the alloy composition and aging temperature as the input of neural network, and the yield strength and fracture toughness as the output. The simulated annealing algorithm was used to optimize the properties of steel and determine the optimal composition and heat treatment process of high-strength and high-toughness steel. Malinov et al. [275] established a model of the

relationship between process parameters and mechanical properties of titanium alloys using neural networks. The input is the working temperature, heat treatment parameters and alloy composition, and the output is the nine most important mechanical properties such as yield strength, impact strength and elongation, among others. The network can not only predict the mechanical properties of titanium alloys, but also optimize the heat treatment parameters and composition of titanium alloys. As shown in Fig. 14, Shariati et al. [276] proposed an exceptional machine learning framework, which enables the concept of determining material properties (i.e., polar diagram) to be associated with processing parameters (i.e., temperature and deformation) without understanding the prior dependence of physical variables. In general, the framework is beneficial for those processing parameters, which a mathematical basis can represent in interpolation. Guo et al. [277] established an artificial neural network (ANN) model to analyze and simulate the correlation between the properties of maraging steel and its composition, working conditions and process. The input parameters of the model include alloy composition, working temperature and process parameters (including cold deformation degree, aging time and aging temperature). The output of ANN model includes a variety of properties, such as elongation, area shrinkage, and fracture toughness, among others. The neural network model achieved good performance. Chaudry et al. [278] established the hardness prediction model of aluminum alloy

under various aging treatment conditions by using 1592 hardness data samples of Al-Cu-Mg-x ($x = \text{Zn, Zr, etc.}$) alloy and taking material parameters such as composition, aging temperature and aging time as inputs. The determination coefficient and mean square error of gradient boosting tree model reached 0.94 and 7.27 respectively, which accurately predicted the aging properties of al-4cu-0.5 mg-0.15si-0.1sc alloy at 175 and 225 °C. Zhao et al. [279] proposed a combination of a machine learning surrogate model based on Gaussian process regression and a phase diagram to find components with better target properties in Cu-Co-Si alloys. An ML model trained from "discarded" experimental data with poor hardness or conductivity establishes a relationship between input variables (composition and heat treatment parameters) and target properties (conductivity and hardness). After five iteration cycles, the highest hardness and conductivity of the alloy were optimized to 207.39 HV and 69.24% IACS, which were 14.7% and 20.8% higher than the best values in the initial training data.

3.6. Service behavior evaluation

The service behavior of materials is closely related to their microstructure, composition and environmental conditions, and the interaction between different factors is complex. Machine learning can obtain the influence law between various factors from a large number of service

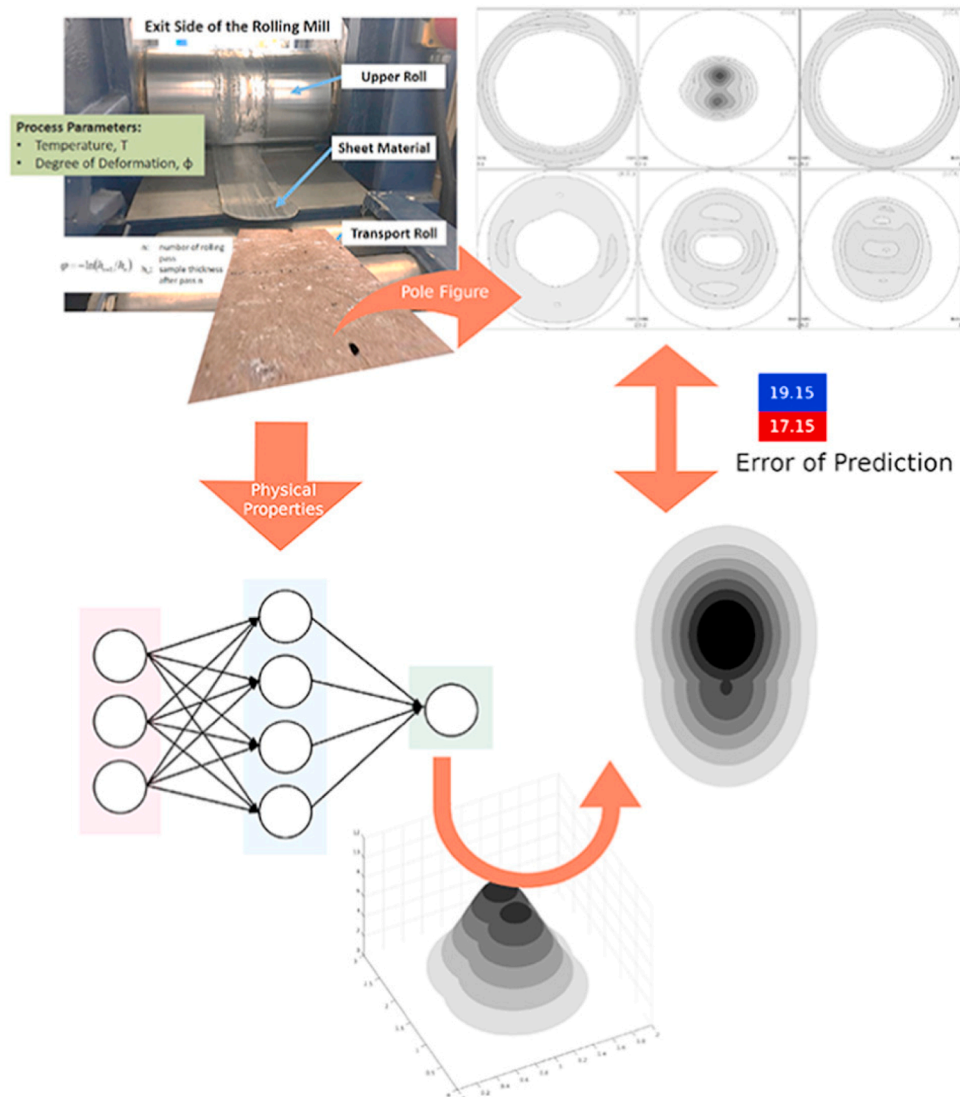


Fig. 14. Enabling intelligent Mg-sheet processing utilizing efficient machine-learning algorithm. Reproduced with permission [276] Copyright 2020 Elsevier.

data, and predict the service properties. The service properties of materials mainly include fatigue, corrosion, wear, creep and irradiation energy. The evaluation of material service behavior belongs to supervised learning, and the commonly used machine learning models include support vector machine, artificial neural network, decision tree and random forest.

Fatigue refers to a phenomenon in which the cracks or fractures occur in structures under the action of alternating loads, which is usually expressed by fatigue life or fatigue strength [280]. When there are few material fatigue samples, overfitting and local minimum are prone to occur, and the prediction accuracy of the model is greatly reduced. Support vector machine (SVM) can effectively avoid over fitting and local minimum problems. Therefore, it is suitable for data sets with small data samples and medium feature dimensions. Wu et al. [281] established a material fatigue life prediction model based on support vector machine. The model prediction error is 7.91%, which can achieve material fatigue life prediction under the condition of fewer samples. However, SVM is easy to fall into local optimum in the process of parameter optimization. Evolutionary algorithm is usually used for optimization to improve the prediction accuracy. Ma et al. [282] established a fatigue life prediction model of ductile iron based on SVM, and the simulation experiment of optimizing SVM is carried out by using genetic algorithm (GA), ant colony optimization (ACO) and partial swarm optimization (PSO). The results show that the prediction effect of DE-SVM algorithm is better.

The factors of material corrosion include chemical composition, microstructure, processing technology, service conditions, etc., and different factors contain multiple dimensional information. Generally, models with strong fitting ability and high applicable dimensions such as artificial neural network and support vector machine are used to predict the corrosion properties. Wang et al. [283] used both alloying elements and environmental factors as input factors, and constructed a complex ANN model with a topology of 13–19–1 to predict the corrosion rate of low-alloy steel. The fitting degree of the prediction model can reach 0.9215, which confirms the generalization ability of the model. Fu et al. [284] used the SVM model to predict the corrosion rate after feature selection of factors affecting the atmospheric corrosion of steel, and compared with the traditional least squares (LS) method to determine the effectiveness of the model to small sample data. Later, Qiu et al. [285] further optimized the corrosion prediction model by combining recursive feature elimination (RFE) and SVM, and the model accuracy and generalization ability were significantly improved. Smets et al. [286] used artificial neural network to describe the stress corrosion of austenitic stainless steel under different temperature, chloride ion concentration and oxygen content.

Wear is related to material hardness, stress state, friction coefficient and other factors, but these parameters are usually difficult to obtain accurately, which is easy to cause data missing. The random forest model with good tolerance for missing values is usually used for prediction. Zhao et al. [287] proposed an evaluation model based on the combination of random forest and principal component analysis to assess the wear status of the cutting tools. Firstly, the key features of tool wear is obtained by dimensionality reduction, and then the wear status of the cutting tools is evaluated. The evaluation accuracy of tool wear state by the model is 94.3%. Dimensionality reduction and important feature screening improve the efficiency and evaluation accuracy of the model, reduce the impact of noise on the evaluation results, and enhance the robustness of the model.

The high cost of creep testing limits the efficiency of developing new alloys using traditional trial-and-error methods. In addition, complex mechanisms and influencing factors significantly increase the difficulty of physical modeling and simulation to guide the design. Machine learning assisted creep property prediction accelerates the design efficiency of such alloys. Xiang et al. [224] used textual (chemical composition and creep test conditions) and visual (as cast microstructure) data to quantitatively predict the creep rupture time of Fe-Cr-Ni

heat-resistant alloys. convolutional neural network (CNN) and Multi-layer perceptron (MLP) are employed to analyze visual and textual data, respectively. Then, the multi-source heterogeneous data are integrated by integrating the deep learning model. Compared with the single CNN and MLP model, the prediction accuracy is significantly improved. Based on the published experimental data, Brun et al. [288] developed a neural network model, which allowed the estimation of creep rupture strength of martensitic and bainitic steels as a function of chemical composition, heat treatment time and temperature. This model, together with various kinetic and thermodynamic calculations, has been used to propose two new alloys related to 10CrMoW steel. For the prediction of material hardness under irradiation environment, Kemp et al. [289] proposed an artificial neural network model based on Bayesian framework, and analyzed the effects of temperature, irradiation dose and chemical composition on the yield stress of irradiated martensitic steel. The neural network model can well capture the nonlinear relationship between yield stress, chemical composition and irradiation conditions, and the prediction results are helpful to the design of material properties test and the supplement of missing information.

3.7. Improve computational simulation technology

The interdisciplinary application of materials science and computer science has promoted the first computational revolution of materials science. Traditional computational simulation methods include density functional theory (DFT), Monte Carlo method (MC), molecular dynamics (MD) and finite element analysis (FEA). Through theoretical calculation and simulation technology, researchers can more effectively explore more microscopic phase and component space, and deal with more complex systems with the help of abstract energy concepts. However, the applicable conditions of each calculation and simulation method are limited. For example, the simulation scale of density functional theory is small, and the simulation results have large errors in the environment of high temperature, high pressure and strong magnetic field; Molecular dynamics simulation depends on more accurate potential function; Finite element simulation requires accurate boundary condition setting, and the accuracy fluctuates greatly. It is a new research idea in the field of computational materials to combine machine learning methods with computational simulation methods and give full play to the ability of machine learning to process big data and extension.

There are many methods to combine material calculation and simulation methods with machine learning. The simplest method is to use the calculation results of calculation and simulation as the training data of machine learning. Meredig et al. [290] used the calculation results in the existing DFT database to establish a forward model trained by DFT data (including structure information) to predict material properties (such as formation energy). The formation energy of new materials can be predicted without inputting crystal structure information. Park et al. [291] used the data calculated by density functional theory to build a machine learning model to predict the stability of organic-inorganic hybrid perovskite materials, which provides an efficient method for material design. Lu et al. [292] obtained the decomposition energy data of halogenated double perovskite through high-throughput density functional theory calculation, and built a machine learning model to predict the thermodynamic stability of these compounds. Li et al. [293] calculated the data containing decomposition energy through high-throughput density functional theory (DFT), and trained a machine learning (ML) model to describe the underlying relationship between structural, chemical characteristics and decomposition energy. Yuan et al. [294] used the first principle calculation data of BaTiO₃-based piezoelectric materials to establish a machine learning model to guide the design and synthesis of BaTiO₃-based piezoelectric materials with large electrostriction, which accelerated the discovery of new materials. As shown in Fig. 15, Zeng et al. [295] first generated the data for more than 300,000 equilibrium phases by

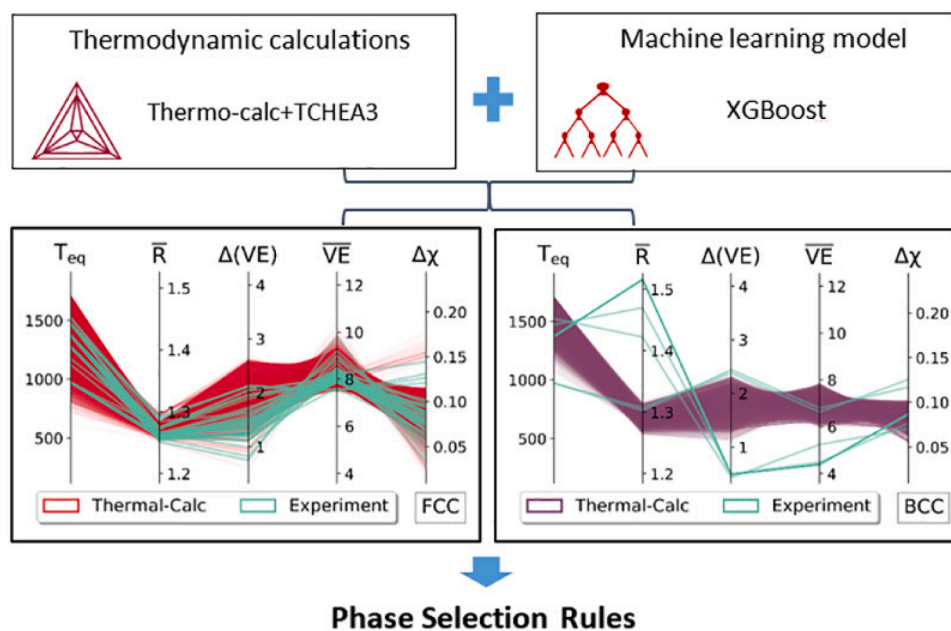


Fig. 15. Revealing high-fidelity phase selection rules for high entropy alloys: a combined CALPHAD and machine learning study. Reproduced with permission [295] Copyright 2021 Elsevier.

thermodynamic calculation, and selected 15 properties as descriptors. Then, the machine learning method is used to identify the five most important descriptors. Based on these five important descriptors, a new phase selection rule for FCC phase and BCC phase is established, with a success rate of more than 90%, which is much better than the phase selection rule constructed based on empirical method. Trowsdale et al. [296] proposed a method combining artificial neural network technology and finite element technology. Firstly, the traditional finite element method is used to calculate the stress and strain, and then the neural network is trained with the obtained data. The "learned" neural network can immediately and accurately predict output values such as stress, strain, deformation, etc., and no longer requires complicated calculations, which not only improves the response speed, but also saves the CPU time. Jung et al. [297] proposed a computational aided material design method composed of Gaussian regression model and phase field simulation technology. This method combines experimental data and simulation data, establishes a model between alloy microstructure and properties, and finds the optimal microstructure through global optimization and methods based on expected improvement.

Machine learning can not only use the calculation results of the computational simulation method to train the predictive properties, but also optimize the calculation process of the computational simulation method itself. Among the existing theoretical calculation methods, various semi-empirical parameters can be trained and summarized by machine learning method, and the flexibility of machine learning can be used to expand the application scope of existing methods. In addition to the density functional [298–302] in density functional theory, the interatomic potential [303,304] and force field [305–308] in molecular dynamics can be fitted by machine learning method.

Density functional theory (DFT) is the standard formalism to study the electronic structure of matter at the atomic scale. In Kohn sham DFT simulation, the balance between computational cost and accuracy depends on the choice of exchange and related functional, which exists only in approximate form. Here, Dick et al. [309] proposed a framework for creating density functional using supervised machine learning, which is called neuralxc. These machine-learned functionals are designed to improve the accuracy of baseline functionals towards that provided by more accurate methods while maintaining their efficiency. Schmidt et al. [310] trained the neural network as a universal exchange

correlation functional of density functional theory, which simultaneously reproduced accurate potential energies and exchange-correlation energies. This function is very non-local, but retains the computational scaling of traditional local or semi-local approximations. Therefore, it is expected to solve some delocalization problems that plague the density functional theory.

As shown in Fig. 16, Podryabinkin et al. [311] proposed an active learning approach for fitting the machine learning interatomic potentials, which is based on the D-optimality criterion and is used to select the atomic configuration suitable for potential energy. The results show that the proposed active learning approach is very effective in terms of dynamic training potential. Mortazavi et al. [312] first employed machine-learning interatomic potential (MLIP) to explore the failure behavior, mechanical properties, thermal expansion and dynamic stability of biphenylene monolayer. By comparing the failure mechanism and the predicted direction-dependent uniaxial stress-strain relationship of biphenylene monolayer with the results obtained by density functional theory simulation, the remarkable accuracy of the developed MLIP-model is obtained.

Chmiela et al. [313] developed an effective gradient-domain machine learning (GDML) approach to construct an accurate molecular force field using a limited number of samples from ab initio molecular dynamics (AIMD) trajectories. The GDML implementation can be trained using only 1000 conformational geometries to reproduce the global potential energy surface of intermediate-sized molecules with an energy accuracy of 1 kcal mol⁻¹ for atomic forces and 0.3 kcal mol⁻¹ for energies. Xu et al. [314] developed machine learning force fields for amorphous and crystalline Li–Si systems with Li/Si ratio ranging from 0 to 4.2. The volume change during the initial liquefaction process was predicted by force-field driven molecular simulations, which was in agreement with the experimental results. The authors also report that the force field can quickly and well predict the bulk density, radial distribution function and diffusion coefficient of lithium in amorphous lithium silicon system. The accuracy is close to that of quantum mechanical calculation, and the speed is 20 times faster than that of quantum mechanical MD simulation.

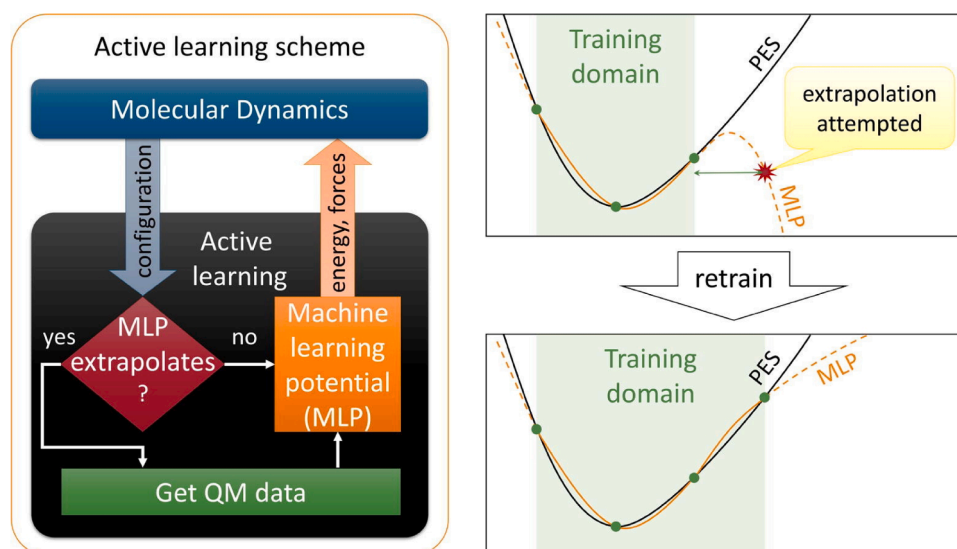


Fig. 16. Active learning of linearly parametrized interatomic potentials. Reproduced with permission [311] Copyright 2017 Elsevier.

3.8. Multiobjective optimization

In the multi-objective optimization problem, each objective cannot be optimized simultaneously, and the sub-objectives are contradictory (such as the strength and toughness of steel materials [315], the creep strength and oxidation resistance of superalloys [316], the hardness and electrical conductivity of copper alloys [317], etc.), the improvement of one target may cause the performance of another or several other objectives to decrease, that is, it is difficult to achieve the optimum of each objective simultaneously. Multi-objective optimization has always been an important research topic in the field of engineering, and it is also a key problem that needs to be solved in the development and application of materials. Machine learning methods combined with multi-objective optimization selection provide a new method for the rapid design and optimization of materials with multi-objective properties [318]. Under the background of data-driven approaches, two typical strategies can be considered for the optimal design of materials with multi-objective properties.

The first method is the traditional optimization scheme, that is, the multi-objective optimization is converted into single-objective optimization by means of weighting coefficients, and then the single-objective optimization problem is solved to solve the multi-objective optimization problem, and finally an optimal solution is obtained. Commonly used conversion methods include weighted sum method [319], ϵ constraint method [320], maximum and minimum method [321], target vector method [322], distance function method [323] and so on.

Chen et al. [324] used the distance function method to convert the double targets into a single target in the European space by setting the expected target properties of the yield strength and elongation of the magnesium alloy in advance, and then used machine learning to predict the mechanical properties of the alloy under different heat treatment process parameters. Combined with the iterative optimization method, through only four cyclic iterations of the heat treatment experiment, the strength and plasticity of the magnesium alloy were increased by 27% and 13.5%, respectively. Yamawaki et al. [325] took the thermoelectric material quality factor ($ZT = R_{th}S^2T/R_{el}$) constructed by using thermal resistance (R_{th}), resistance (R_{el}), Seebeck coefficient (S) and temperature (T) as the target, used the calculation data of multi-functional alternating transmission of phonons and electrons, optimized the structure of graphene nanoribbons (GNRs) with potential high thermoelectric properties through Bayesian optimization, and developed a new thermoelectric material with a thermoelectric quality factor 11 times higher than the original material. Diao et al. [326] identified a set of mutually

exclusive properties (tensile strength and elongation) and key descriptors of corresponding properties. All key descriptors are regarded as input features of comprehensive performance (CP), which is calculated from the product of tensile strength and elongation. Finally, we developed a machine learning prediction model for CP and successfully applied an efficient global optimization algorithm to optimize two mutually exclusive mechanical properties.

The advantages of these traditional optimization methods are solid theoretical foundation, few random factors and stable optimization effect. However, this method has two disadvantages: (1) each iteration optimization can only get one optimal solution, so it needs to run the algorithm many times; (2) each objective is given a weight, which requires the decision-maker to pre-select and understand the preference relationship between objectives, which is obviously very difficult.

The second method is the intelligent optimization scheme, that is, the evolutionary algorithms such as GA, MOGA, NSGA, NSGA-II are used to directly solve the problem, and a subset containing a large number of non-inferior solutions, namely the Pareto solution set, is obtained, and then the optimal solution is selected from multiple Pareto solutions according to the engineering needs [327]. The second method directly adopts the evolutionary algorithm to solve the problem, and can obtain a subset with a large number of non-inferior solutions, which has stronger search ability than the traditional optimization method that can only obtain one optimal solution.

Menou et al. [328] combined genetic algorithm, Gaussian regression and thermodynamic calculation to design the composition of nickel-based superalloys based on multi-objective performance requirements such as high creep rupture strength, low density and high microstructure stability. The weight percentages of Fe, Co and Ni, ball-to-powder ratio, grinding speed, grinding time, annealing temperature and annealing time were selected as input parameters. Meanwhile, the magnetization saturation (μ_s), coercivity (h_c) and grain size of Fe-Co-Ni nanostructured alloys were considered as output parameters. In addition, Menou et al. [329] used the multi-objective optimization genetic algorithm combining thermodynamic modeling (CALPHAD), solid solution hardening (SSH) and data mining to design high entropy alloys (HEA). The approach searches for the best compromise between SSH, density and single-phase stability. Thousands of Pareto-optimal base-centered cubic (BCC) HEAs are designed. Zeraati et al. [330] employed multi-objective artificial neural network (ANN) and genetic algorithm (GA) to optimize and improve the magnetic and structural properties of Fe-Co-Ni nanostructured ternary alloys. Lu et al. [331] proposed a hybrid model combining a multi-objective gray wolf

optimizer and a support vector machine to predict composite properties in six datasets. The results show that the model performs well in the property prediction of composite materials, and the mean absolute percentage error of the prediction ranges from 0.14% to 5.574%. Based on Pareto optimal solution and NSGA-II algorithm, Peng et al. [332] proposed an improved NSGA-II algorithm, called NSGA-II with preference (NSGA-IIP). When NSGA-IIP algorithm calculates the crowding distance, the cosine similarity is introduced to make the next generation population gather in the direction of preference. This method is applied to the material design of a thermal barrier ceramic coating. From the results, it is shown that the proposed NSGA-IIP achieves better practicality and temporal performance for new material design. Mahfouf et al. [333] focused on the design of heat treatment regimen and the weight percentage of steel chemical composition, defined mechanical properties such as tensile strength and reduction of area as objectives, and used evolutionary multiobjective (EMO) optimization algorithms to determine the required optimal alloy composition and heat treatment temperature. For the multi-objective optimization of ductility and strength of low-carbon ferrite-pearlite steels, Ganguly et al. [334] made a comparative study between the multi-objective genetic algorithm NSGA-II and the conventional goal attainment strategy. After finding that genetic algorithm is more suitable for such problems, Pareto frontier was developed, which gives useful ductility and strength ranges in alloy design. As shown in Fig. 17, Solomou et al. [335] developed a multi-objective material discovery framework based on Bayesian approaches. The capabilities of this framework is demonstrated in an example case related to the discovery of precipitation strengthened NiTi shape memory alloy, which has up to three required properties. Hu et al. [336] proposed an artificial neural network model combined with genetic algorithm (GA) optimization to design the composition of nickel base superalloy. The model can finally find the global optimization values of yield strength and creep fracture life. A set of nickel base superalloy compositions were designed with more than 90% real accuracy to meet the requirements of microstructure and properties. In order to find an alloy suitable for high temperature structural applications, the creep rupture stress, yield stress and ultimate tensile strength are maximized, while the γ - γ' mismatch parameter and brittleness

temperature range are minimized. Menou et al. [337] introduced a new computational framework for optimal design of alloy system. Based on a multi-objective genetic algorithm, 5669 alloys are predicted in Pareto front. A cheaper and easier to weld alternative than Inconel 740 H (In740H) and Haynes 282 was proposed.

In addition to guiding the composition design of materials, the multi-objective optimization method can also be used for the optimization of material processing technology. Li et al. [338] established a multi-objective optimization mathematical model for the thermal deformation process parameters of the alloy by taking the thermal deformation process parameters such as deformation temperature, deformation degree and deformation rate as design variables, and took the optimal synthesis of microstructure and forming force as the goal. The artificial neural network prediction model of microstructure parameters and forming force is used as the knowledge source of the optimization algorithm, and the artificial neural network is combined with the modified genetic algorithm to optimize the thermoplastic forming process parameters of Ti-15-3 alloy. Zoghipour et al. [339] used genetic algorithm-based analysis (GA) and desirability function (DF) for various cutting responses (e.g. dimensional accuracy (DA), metal removal rate (MRR), cutting forces (Fr) and machining time (tm)) for multi-objective optimization. Considering industrial and technological constrain, optimization has yielded minimal dimensional accuracy errors, cutting forces, production times, and maximum material removal rates. Abidi et al. [340] employed a multi-objective genetic algorithm (MOGA-II) to analyze and optimize micro electrical discharge machining (μ EDM) process. The objectives include determining the best machining parameters to obtain better good surface finish and material removal rate. The results indicate that MOGA-II is an effective tool to optimize input parameters. Zhou et al. [341] used the multi-objective genetic algorithm NSGA-II to optimize the response of process parameters, and obtained the Pareto solution corresponding to the combination of stamping speed and blank holder force. Finally, the strength and toughness of the forging can be effectively improved and the material utilization rate can be improved by the stamping test according to the optimal process parameters. Zhang et al. [342] combined the Taguchi method with finite element method to optimize the precision forming

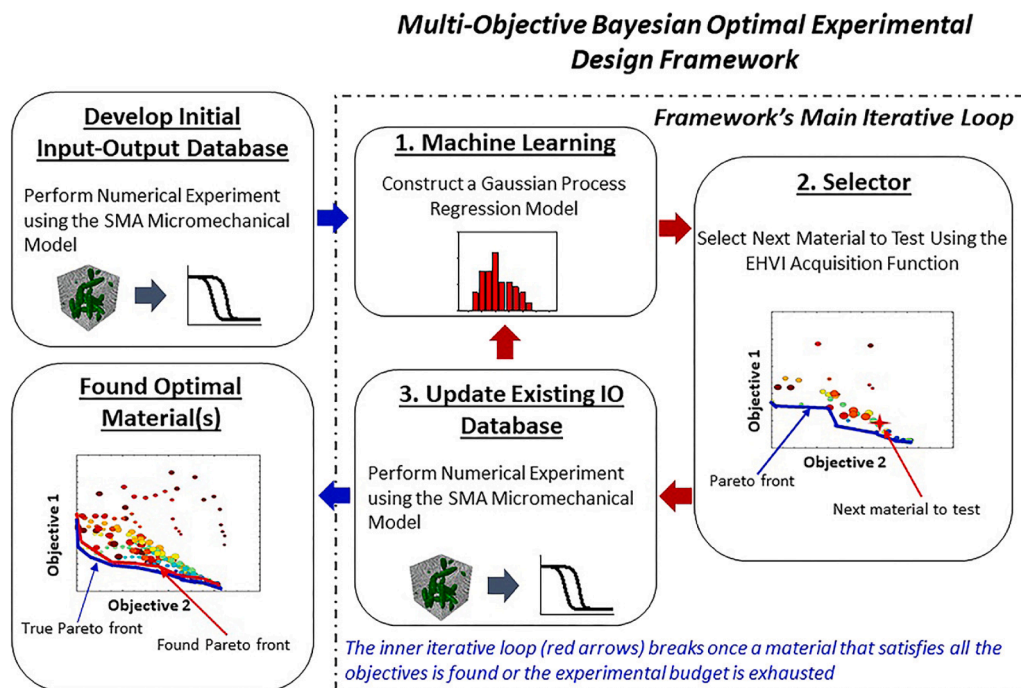


Fig. 17. Multi-objective Bayesian materials discovery: Application on the discovery of precipitation strengthened NiTi shape memory alloys through micro-mechanical modeling. Reproduced with permission [335] Copyright 2018 Elsevier.

process of rib web forgings through multi-objective design. The micro and macro structure of the product were used to verify the simulation results, and the signal-to-noise ratio analysis and variance analysis were carried out on the simulation results to obtain the optimal parameter combination of the analysis process window. The multi-objective optimization technique was used to find the optimal setting of milling parameters for Inconel 718 Alloy. Senthilkumar et al. [343] studied the effects of various machining parameters such as cutting feed, speed, and depth on surface roughness, MRR and tool wear. Three different coated carbide inserts are used in the machining process and perform multi-objective optimization to provide minimum surface roughness and tool wear and maximum MRR. Pitchipoo et al. [344] proposed a multi-objective optimization technology based on the Dragonfly algorithm (DFA) to optimize the weld strength of the aluminum alloy 6082-T6 friction stir welding. This work mainly aims to determine the optimum parameter combination of input process parameters, so as to achieve the optimum impact strength and tensile strength of the weld. Kamal Babu et al. [345] used artificial neural network (ANN) to model the relationship between the input parameters of welded joints and the corrosion and mechanical properties (output), and used genetic algorithm (GA) to determine the optimal friction stir welding parameters to obtain a defect free welded joint with the maximum welding strength. Arunvikram et al. [346] studied the optimal multi response evaluation of process parameters (cutting depth, feed rate and tool speed) using Taguchi based Dragonfly algorithm. Finally, the optimal output characteristics of tool vibration displacement amplitude, surface roughness and hardness are obtained. In order to optimize the sheet metal forming process, Li et al. [347] used the multi-objective genetic algorithm based on Pareto. The optimal model proposed optimizes draw-bead restraining force and blank-holding force as design variables, so that objective functions of thickness change, insufficient stretching, wrinkling, and fracture are minimized simultaneously. In order to optimally balance ductility, hardness and strength, Ghiabakloo et al. [348] determined the annealing conditions of severely deformed low-carbon steel plates by adjusting the annealing temperature and time. Due to the conflicting nature of the required metal properties, a set of optimal annealing conditions was determined by Pareto multi-objective optimization.

3.9. Inverse design materials

Based on the existing data, the machine learning model is established to predict the material properties to guide its composition design, which belongs to the forward design of materials [349]. Due to the huge feature space of candidate materials, the forward design that tries to cover all candidate material feature spaces requires a lot of data. Although the optimization algorithm is integrated, it is difficult to ensure the global optimization of the feature space of materials. It is the ultimate goal of material science research to achieve the inverse design of alloy composition, processing technology and structure from the given alloy properties requirements, which will greatly shorten the research and development cycle of new materials and reduce research and development costs [350]. However, due to the complex nonlinear relationship among alloy composition, process, microstructure and properties, it is very difficult to achieve this goal, and machine learning is accelerating the realization of this goal [351]. The following are examples of commonly used strategies for material inverse design, including deep learning strategies, evolutionary algorithm strategies, generative adversarial network strategies, variational autoencoder strategies, and Gaussian mixture regression strategies.

Tagade et al. [352] proposed a deep learning inverse prediction framework of "Structure Learning for Attribute-driven Materials Design Using Novel Conditional Sampling (SLAMDUNCS)", which successfully designed organic molecules with required properties by combining Bayesian inference with deep learning network. This framework can be easily extended to the design of inorganic materials, such as semiconductors, piezoelectric and thermoelectric materials. Wang et al.

[353] compared the reverse design method of gold nanoparticles based on the neural network for error back propagation (BP) training and the least square method. The results show that the proposed inverse design method based on deep learning can change the mathematical model of metal nanoparticles according to the actual needs, which is suitable for the inverse design of more nanoparticle materials, structures and morphology. Jiang et al. [354] employed the deep learning method to establish the mapping between the dispersion relationship and the structural topology of elastic metamaterials. The results show that the proposed model can accurately predict the dispersion relation of a given structure and the inverse design of near-optimal structures based on the target dispersion relation.

In practical engineering, the inverse design from property to microstructure is more urgent than the regular design from microstructure to property. Chang et al. [355] developed a data-driven machine learning (ML) model based on the combination of genetic algorithm (GA) and artificial back-propagation neural network (BPNN), which is used to design the auxetic metamaterial with specific Poisson's ratio (i.e. zero Poisson's ratio). Mi et al. [356] used the particle swarm optimization (PSO) algorithm to optimize the input of artificial neural network (ANN), thus achieving the "inverse design" from "target performance" to "combination and process". Four kinds of magnesium alloys with better performance were proposed by a machine learning inverse design model. Dong et al. [357] proposed a transfer learning based inverse optical materials design algorithm for suggesting material compositions with a desired target light absorption spectrum. The method is based on the combination of deep neural network model and global optimization algorithms including a Bayesian optimization and genetic algorithm. Zhang et al. [358] developed a new software package with novel algorithm, called IM2ODE (inverse design of materials by Multi-objective differential evolution). The target properties of interest include the hardness of crystals, the electronic structure and optical properties of semiconductors, and so on. IM2ODE can easily predict the atomic configuration with the required characteristics for three-dimensional structures, clusters and interfaces, and even complex defects in solids.

Structural exploration and design optimization to achieve excellent mechanical properties through inverse machine learning is a new area of concern. In order to explore and optimize the structural design such as lightweight lattice unit cells, Challapalli et al. [359] proposed an inverse design framework using generative adversarial network (GANs). The proposed framework finds different lattice unit cells, and their carrying capacity is 40–120% higher than that of the octet unit cells. Dan et al. [360] proposed a generative machine learning model (MatGAN) based on a generative adversarial network (GAN) to inverse design inorganic materials. Using the materials in the ICSD database for training, the GAN model can generate hypothetical materials that do not exist in the training data set, reaching a novelty of 92.53% when generating 2 million samples. A forward analytical framework for predicting the properties of polymer alloy phase separation structures has been developed by Hiraide et al. [361] and the structures have been generated from these properties through inverse design. Convolutional neural network (CNN) performs the forward analysis, and the inverse design is performed by random search towards a model combining a CNN and generative adversarial network (GAN). As shown in Fig. 18, Long et al. [362] developed a inverse design framework, namely constrained crystal depth convolution generation adversarial network (CCDCGAN), which can automatically generate stable multi-component crystal structures. Their formation energy can be optimized in the latent space based on reversible crystal images with continuous representation. Dong et al. [363] proposed a novel regression and conditional generative adversarial network (RCGAN) for inverse design of representative two-dimensional materials, boron-nitride (BN) hybrids and graphene. RCGAN can autonomously generate BN hybrids/graphene given any target bandgap values.

The ultimate goal of materials research is the non-accidental

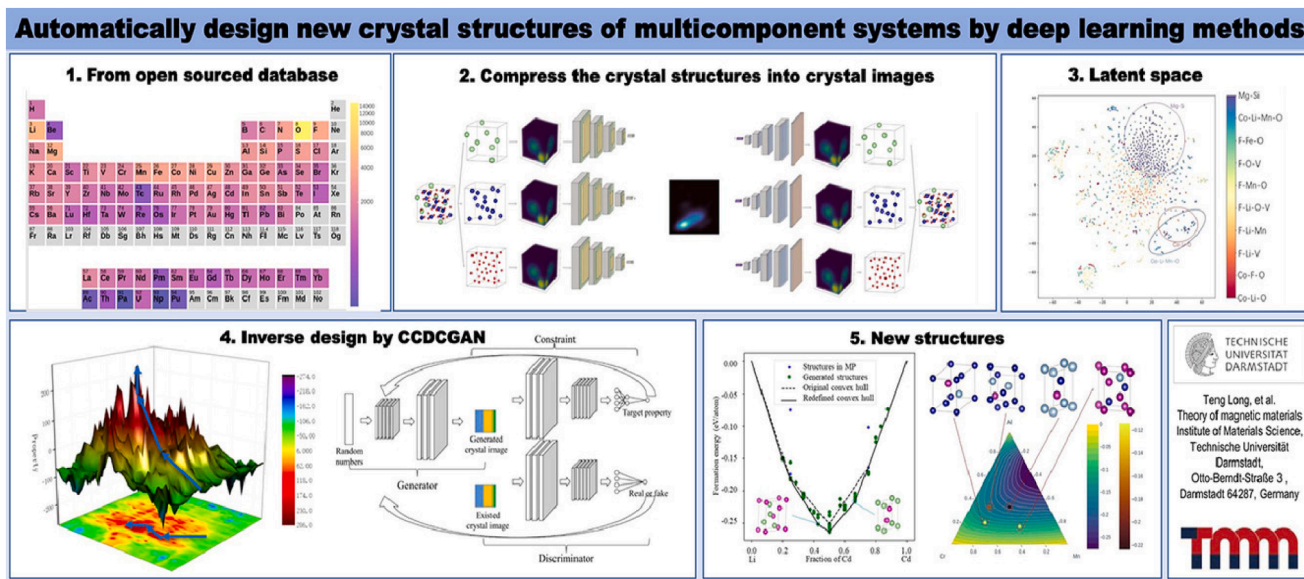


Fig. 18. Inverse design of crystal structures for multicomponent systems. Reproduced with permission [362] Copyright 2022 Elsevier.

discovery of materials with targeted properties. Noh et al. [364] proposed a framework of variational autoencoder strategy for machine learning inverse design of inorganic functional materials. The ability of the autoencoder to generate experimental material is demonstrated by rediscovering experimentally known structures. About 20,000 hypothetical materials have been produced, resulting in several new metastable materials that can be synthesized. The comparison with genetic algorithm shows that the computational efficiency of the generated model is very high, and the chemical composition space can be effectively explored by learning the distribution of known materials, so as to predict the crystal structure. Kim et al. [365] employed a variational autoencoder (VAE) to generate a continuous microstructure space based on synthetic microstructural images. Gaussian process regression (GPR) was used to establish the relationship between the latent space point and

the ferrite grain size as inputs and mechanical properties as outputs. GPR with VAE successfully predicts the newly generated microstructures with target mechanical properties with high accuracy.

As shown in Fig. 19, Shimizu et al. [366] focused on Gaussian mixture regression (GMR), which can simultaneously handle multiple Y variables. GMR can predict multiple Y variables at the same time and consider the relationship between Y variables. It can also directly predict X variables by substituting the values of multiple Y variables. Based on the advantages of forward and inverse analysis of GMR model, the data sets of thermoelectric conversion materials are analyzed, and new high-performance thermoelectric conversion materials are sought. Although many approaches actually use pseudo-inverse analysis, Gaussian mixture regression (GMR) can achieve direct inverse analysis. Kaneko et al. [367] described the development and use of extended GMR

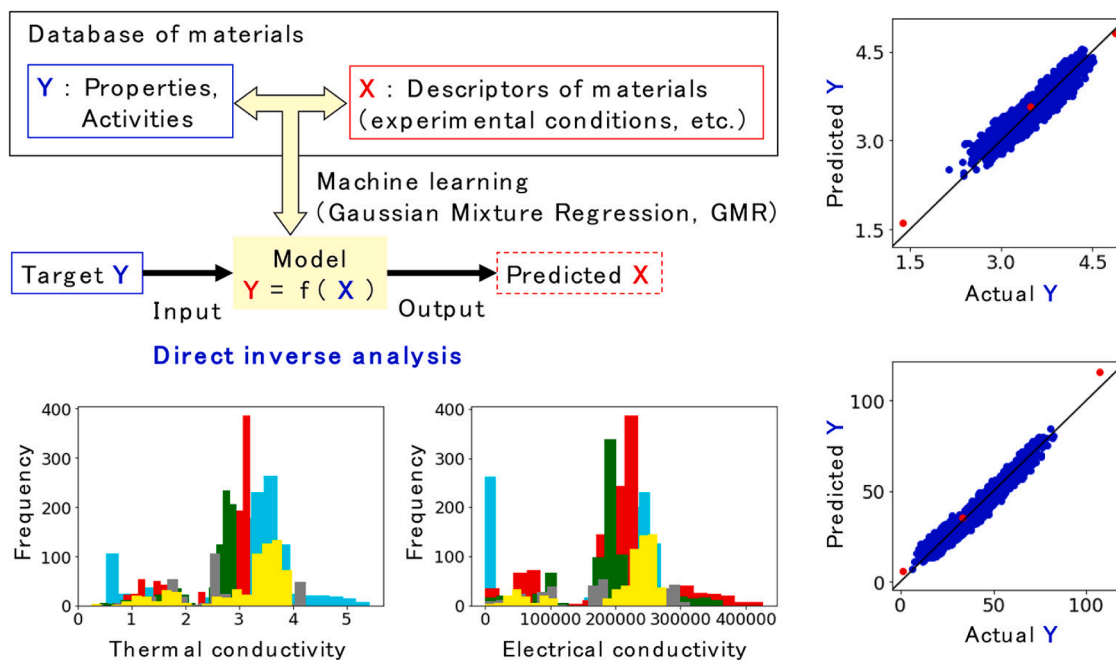


Fig. 19. Direct inverse analysis based on Gaussian mixture regression for multiple objective variables in material design. Reproduced with permission [366] Copyright 2020 Elsevier.

(EGMR), which provides higher prediction ability than traditional GMR. Compound datasets, numerically simulated datasets, spectral datasets, and a material dataset are used to verify the effectiveness of the proposed EGMR.

Of course, in addition to the inverse design strategies mentioned above, some other inverse design strategies or method frameworks (inverse design expert systems [368,369], back-propagation optimization strategies [370], topology optimization methods [371,372], Other reverse design frameworks [373–375]) are also significantly accelerating the development of the field of inverse material design. Due to space limitations, they will not be presented here.

4. Common problems and countermeasures

4.1. High dimensionality and small sample

The high dimensionality of datasets and limited number of samples are common challenges in machine learning. The higher dimension of data cause information redundancy and increase the amount of calculation required; the lack of data may lead to a variety of consequences: including overfitting, difficulty in proving the reliability, generality and applicability of the prediction model. In order to reduce the impact of high-dimensional and small-size data on the prediction accuracy of the model, dimensionality reduction methods [376,377], active learning [378,379], data enhancement [380], model generation [381], meta learning [382] and ensemble learning [383] can be used in machine learning.

The dimensionality reduction method can overcome the disaster of dimensionality, obtain essential features, save storage space, remove useless noise and realize data visualization. The four classical dimensionality reduction methods are linear discriminant analysis, principal component analysis, locally linear embedding and laplacian eigenmaps [384,385]. For instance, Hu et al. [386] proposed a tool wear-monitoring strategy. The mutual information was adopted for feature selection. Then, linear discriminant analysis was used to reduce the dimension and find the best training data set. Finally, ν -Support vector machine was applied for training and prediction. The prediction accuracy of this method is 98.9%, which can be used for tool wear monitoring. Bundela et al. [158] trained and tested eight different machine learning algorithms to predict microhardness. The accuracy of prediction improved by reducing the higher-dimensional data to lower dimensions using principal component analysis. Current research shows that the test R2 score of Bagging regressor, Random forest, and XGBoost algorithms exceeds 0.89. In order to extract features related to welding quality and reveal the effectiveness of automatic feature extraction method based on dimension reduction technology, Zhao et al. [387] used local linear embedding (LLE), isometric mapping (Isomap) and principal component analysis (PCA) approaches to process dynamic resistance signals of different welding process parameters. The results show that the regression model based on the features extracted by LLE technology is more robust. Harefa et al. [388] utilized support vector machine classifier model and manifold dimensionality reduction technology to quickly classify five different types of aluminum alloys. In order to circumvent the curse of dimensionality, nonlinear manifold learning techniques, such as local linear embedding (LLE), local tangent space alignment (LTSA), Laplacian eigenmaps (LE), and isometric mapping (Isomap) were used. Compared with nonlinear techniques, the performance of linear techniques such as multidimensional scaling (MDS) and principal component analysis (PCA) was also studied. The prediction labels indicated that Isomap-SVM model has the best classification performance. In addition, other traditional or novel dimensionality reduction methods can also refer to some review reviews [389–394].

Active learning is a way to construct an effective training set. Relying on this method, the learner can actively select training data, and the learner will reduce the demand for a large number of labeled samples.

For example, Gubaev et al. [395] adopted an active learning algorithm to select the best training dataset automatically. This method greatly reduces the density functional theory calculation and can generate training data only by DFT. Murphy et al. [396] proposed an active learning method for high-dimensional image classification, which uses the nonlinear diffusion geometry of spatial regularization to represent the clustering core. The proposed method samples from the estimated cluster core to generate a small but effective training label set. This method avoids the redundancy of the training dataset, which produce high-precision labels with few training labels. Data enhancement allows limited data to generate more data, including increasing the number and diversity of training samples, increasing noisy data and improving model robustness. Data enhancement methods include mirroring, rotation, migration, scale transformation, random extraction and color jitter [397]. For example, Banerjee et al. [398] sought to further improve the binary classification model for high-throughput identification of different microstructural morphologies. A dataset with limited observations (444 non-dendritic structures, 133 dendritic structures) were utilized and enhanced sixfold by migration and rotation. Then, transfer learning is carried out using pre-trained networks. In addition, more samples can be generated through generative models (such as generative adversarial network [399,400], variational autoencoder [401,402]). For example, in order to meet the requirement of machine learning (ML) method for a large number of experimental data in the multiaxial fatigue life prediction of 316 L stainless steel, Sun et al. [403] proposed a cyclic generation countermeasure network (named cGAN) combined with Fourier transform and other semi-empirical equations to augment data following physical knowledge. Two samples of each loading path can be augmented into hundreds of good quality samples. Li et al. [404] combined discriminative neural networks (NNs) with generative adversarial networks (GANs) to accelerate the exploration of a novel multi-principal element alloys (MPEAs). By applying the GAN model herein, it was possible to directly generate novel compositions for MPEAs, and to predict their phases. As an application in materials science, Ohno et al. [405] took seven benchmark data sets and one ionic conductivity data set as the research object, and used the variational auto-encoders (VAEs) as the generation model of data augmentation to solve the issue of small data size for regression problems. The experimental results show that the multi-task learning (automatic coding and regression tasks) for VAEs improves the generalization performance of multivariable linear regression model trained with augmented data. In addition, VAE is also used to map high-dimensional data of complex microstructure to low-dimensional space [406,407]. Another promising solution to limited datasets is meta learning, that is, learning knowledge within and between problems [408]. Meta learning can achieve the task of small sample classification well. It can learn efficiently in only a small number of samples so that the model can learn well through a few examples. For example, Lee et al. [409] proposed a meta-learning approach to solve the few-shot regression problem. The proposed method is based on data augmentation using optimization-based meta-learning and traditional regression models. The resulting deep neural network can be used to optimize the reactive-sputtering process used to fabricate nitride and titanium composite thin films. There are three ways to realize small sample classification by meta learning: metric-based meta learning, model-based meta learning and optimization-based meta learning [410]. Ensemble learning completes the learning task by constructing and combining multiple machine learners, which is also known as "learning from others". When modeling high-dimensional and small-scale data, ensemble learning can usually achieve higher performance than a single learner [411]. For example, Tsai et al. [412] fully explored the performance of two ensemble learning feature selection techniques (serial and parallel combination methods) for feature selection on low sample size and high-dimensional data. By comparing nine serial and nine parallel combinations and three single baseline feature selection methods, the experimental results show that the integrated feature selection can achieve better classification accuracy than single

feature selection.

4.2. Construction of effective feature descriptor

Feature engineering is the most time-consuming work of data analysis. For different types of materials, there are specific differences in feature types. Data and features determine the upper limit of machine learning, and models and algorithms approach this upper limit. Therefore, feature engineering becomes particularly important. Machine learning uses features to construct models, and its success depends on the validity and relevance of these features. A good feature must be more accessible to obtain than the target attribute and have the lowest dimensionality [413]. In materials research, each feature is usually only for a specific application, and there is no uniform feature that is effective for all applications. Therefore, constructing the most suitable feature for each machine learning process is another challenge.

At present, researchers have done some interesting work in building effective feature descriptors. For example, Chen et al. [414] collected the Charpy impact toughness (CIT) data of low alloy steel and established a CIT prediction model based on machine learning algorithm. Three feature construction strategies are proposed, one is completely based on alloy composition, the other is based on alloy composition and heat treatment parameters, and the last is based on alloy composition, heat treatment parameters and physical features. A hybrid method combining filter and wrapper is used to select relevant features. Compared with the strategy based on alloy composition only, the strategy based on alloy composition, heat treatment parameters and physical features performs better. Zhang et al., [415] first extracted physical and chemical parameters of 79 elements that may affect alloy properties, and used special equations to construct 158 alloy features. Based on domain knowledge, 12 alloy features were added, and finally 170 candidate feature combinations that affect alloy hardness and conductivity were obtained. Then identify a set of key features through relevant screening, recursive elimination and detailed screening of existing data sets, so as to significantly enhance the combined ultimate tensile strength (UTS) and electrical conductivity (EC) of the alloy. Zhang et al. [55] first constructed the material descriptor space by considering element parameters, thermodynamic and kinetic parameters, structural information, density functional theory (DFT) calculations, and specific problem descriptors. Then genetic algorithm (GA) is used to effectively select the best ML model and material descriptor combination from the material descriptor space and ML model pool, and its effectiveness for two-phase formation in high entropy alloy (HEA) is demonstrated. Himanen et al. [416] developed a software package describing machine learning, providing a widespread feature transformation (descriptor) for atomic material simulation. Describe accelerates the application of machine learning in atomic property prediction by providing user-friendly and ready-made descriptor implementation.

In addition to the above means of constructing effective features, automatic feature engineering and deep learning provide two new ideas for constructing effective features. For example, automatic feature engineering has become an emerging topic of academic research [417]. The automatic feature engineering aims to automatically create candidate features from a dataset and select several best features for training. The automated feature engineering helps data scientists shorten the time to explore data, enabling them to try and misrepresent many ideas in a short time. On the other hand, it enables non-experts who are not familiar with data science to extract value from their data quickly. It only takes minimal effort, time and cost. The main automation feature engineering tools include feature tools, AutoFeat and tsfresh. Featuretools uses an algorithm called Deep Feature Synthesis (DFS), which is a framework for performing automated feature engineering [418]. AutoFeat library has two functions: automatic feature derivation and automatic feature filtering. That is, AutoFeat derives nonlinear features through power, logarithm, addition, subtraction, multiplication and division, and automatically filters out valuable features. The feature

derivation function of AutoFeat can be seen as an advanced function of Featuretools. Featuretools realize the derivation from original data to basic variables, and AutoFeat completes the derivation and screening from basic variables to nonlinear variables [419]. The main function of Tsfresh is to extract various signals in the time series as features, which can effectively expand the feature dimension in the time change dimension. The package contains a variety of feature extraction methods and feature selection algorithms [420].

With deep learning, feature engineering is less important, because deep learning provides an alternative method to eliminate features or feature engineering, and omitting the process of feature engineering will significantly accelerate the machine learning process. In reference [421], Dharmadhikari et al. studies two distinct deep neural network (DNN) architectures for fatigue damage detection in aluminum. The first DNN, called as a feature-based network. The second DNN, called as a feature-less network, uses the ultrasonic data as-is without any pre-processing. The feature-less network is observed to outperform the feature-based network with an accuracy of 94.26% and 98.94% for the two stages. In reference [422], Nassar et al. present the idea of using compositional (atomic percentage) data with a simple neural network to predict HEA phases. They also explore a neural network which uses HR features along with the atomic percentage data and compare the two models. The average testing accuracy of the neural network using only component data (NN1) is better than the neural network working with HR features as well as compositional data (NN2). That NN1 shows that the complete abandonment of predictive features is not only possible but also advantageous. In reference [423], a deep learning-based method was developed as an efficient portable tool, which directly skips feature engineering steps and can directly identify the elastic isotropy of structural materials from unit images with arbitrary component distribution. It has achieved good results in isotropic recognition, with an accuracy rate of about 90%, and the processing time for each sample is milliseconds. On the other hand, although deep learning allows us to eliminate the tedious process of feature engineering, it also makes us lose the understanding of features, such as the importance of features.

4.3. The complexity of machine learning

The complexity of applying machine learning in materials science is reflected in two aspects.

(1) hyperparameters optimization is one of the challenges in machine learning algorithm. The purpose of hyperparameters optimization in machine learning is to find the hyperparameters that make machine learning algorithm perform best on the validation dataset. In many cases, engineers rely on trial and error method to manually adjust and optimize the hyperparameters. Experienced engineers can determine how to set the hyperparameters to obtain higher model accuracy. However, this method relies on a lot of experience, and it is time-consuming, and the model prediction accuracy is not high. In order to solve this problem, some hyperparameter optimization methods are used to simplify the complex hyperparameter adjustment process of the model.

1) Grid search is the most basic hyperparameter optimization method. We only need to construct independent models for all possible hyperparameters, evaluate each model's performance, and finally select the model and hyperparameters with the best results [424]. For example, Kaneko et al. [425] proposed a fast optimization method of support vector regression hyperparameters based on grid search (GS) and cross validation (CV) method. After values of two hyperparameters are determined in theory, GS and CV are used to optimize each hyperparameter, respectively. The highly predictive ability of SVR models and small computational time for the proposed method is confirmed through the case studies using real datasets.

2) Generally, not all hyperparameters have the same importance, and some may play a more significant role. A random search can more accurately determine the optimal value of some essential

hyperparameters than the grid search method [426].

3) The first two methods can train individual hyperparameter combination models and evaluate their performance. Each model is independent, so it is easy to perform parallel calculations. But the independent model also lead to the lack of guidance between models, and the calculation results of the previous model cannot affect the hyperparameter selection of the latter model [427]. However, the Bayesian optimization method can learn from the current results and then affect the choice of the superparameters of the model. Guan et al. [428] quantitatively predicted the melting temperature of binary alloys by constructing a machine learning model "MeltNet". The influence of model hyperparameters on prediction accuracy is systematically studied, and the optimal hyperparameters are obtained by Bayesian optimization.

4) Gradient-based optimization methods are often used to calculate the gradient of hyperparameters in neural network models and are optimized by gradient descent algorithms [429].

5) The idea of evolutionary optimization comes from biological concepts. Since natural evolution is a dynamic process in a constantly changing environment, it is suitable for hyperparameter optimization problems because hyperparameter optimization is also a dynamic process [430]. Bdolhossein et al. [431] predicted the density of binary mixtures by combining coupled simulated annealing optimization tool and the least square support vector machine strategy, which simplifies the hyperparameter adjustment process and achieves an RMSE of 1.69%. Zhu et al. [432] adopted the global optimization algorithms to optimize the pattern center and orientation for electron backscatter diffraction patterns as well as deformation state extraction. The mutation schemes and hyperparameter space of differential evolution (DE) algorithm were thoroughly investigated, and it is proved to be a more effective algorithm than particle swarm optimization (PSO).

In addition, there are some hyperparameter optimization methods that can be used for reference: for example, hyperparameter optimization is carried out with strategies such as reinforcement learning [433], multi-objective reinforcement learning [434], feed-forward and back-propagation artificial neural networks [435], and univariate dynamic encoding algorithm [436].

(2) The process of machine learning is very complex, and the demand for expert knowledge limits the application of machine learning methods. In industrial application scenarios, machine learning is usually proved to be superior to traditional white box modeling. However, due to human factors in the development process, the certainty of finding a solution close to the theoretical optimal value is very low. More robust network models have been constructed with the wide application and development of the deep neural network, such as AlexNet, VGGNet, GoogleNet and ResNet [437,438]. Although these models are flexible enough, the artificial neural network still needs a lot of professional knowledge and time, and parameter adjustment is also difficult for the deep model. Whether it is possible to automate this process, so that everyone, even those who do not understand machine learning, can easily apply machine learning to the problems they face, and automated learning (AutoML) is the answer. AutoML allows complete automation from feature extraction, preprocessing, model selection to hyperparameter optimization [439]. For example, a method based on automatic machine learning architecture [440] can train the machine learning and statistical prediction models of the chemical properties of metal organic frameworks and estimate their prediction performance with confidence intervals. The architecture tries to combine several different machine learning algorithms, adjusts their hyperparameters, and conservatively estimates the performance of the final model. The results show that it can correctly estimate the performance even with few samples (< 100), and it can provide better prediction than trying to use a single standard method (such as a random forest). AutoML democratizes ML as non-professional material science practitioners, who may not know which algorithm to use on a given problem, how to adjust them, and how to correctly estimate their prediction performance to

improve productivity and avoid common analysis pitfalls significantly. In order to accurately predict the performance of the tunnel boring machine (TBM), the best combination of a data preprocessing method, a machine learning model and the related hyperparameters is found, and an optimal mean square error value of 3.148×10^{-2} is obtained using AutoML [441]. The prediction results prove that AutoML is a powerful tool that can effectively predict TBM performance and reduce the need for expert knowledge of machine learning.

4.4. Understanding the "black box"

Enhancing the interpretability and transparency of the machine learning model will help to eliminate model errors, guide the direction of data collection in the future, provide real and reliable information for feature construction and human decision-making, and finally establish trust between human and model. Researchers in artificial intelligence have made significant breakthroughs in natural language understanding, image recognition, and chess in recent years. However, due to a lack of understanding of complex machine learning models, the models are skewed in their dissemination. In selecting an appropriate machine learning model, we usually need to consider the relationship between model interpretability and accuracy. Black box models, such as gradient enhancement models, neural networks or complex ensemble models, typically have a high accuracy [442]. The internal working mechanism of these models is however a challenge to understand, and it is impossible to estimate the importance of each feature for the predicted model output, let alone to understand the interplay between different features. On the other hand, simple white box models such as linear regression and decision tree usually have limited prediction ability. It is impossible to model the data with complex relationships in the dataset [443]. However, simple models usually have better interpretability, and the internal working principles are easier to explain. After all, machine learning models are still a "black box" most of the time and can be used in many cases, but the specific principles are not very clear. That is, it lacks sufficient "interpretability". Therefore, how to improve the comprehensibility of the model is a problem that needs to be solved immediately.

The interpretability of machine learning models can be divided into two categories [444,445]: ante-hoc interpretability and post-hoc interpretability. Ante-hoc interpretability refers to the ability of the model itself to be interpretable by training a model with a simple structure and good interpretability or a self-explanatory model that combines interpretability into a specific model structure. Classical ante-hoc interpretation methods include self-explanatory models [446], generalized additive models [447] and attention mechanisms [448]. For example, based on multiple linear regression, polynomial regression and other self-explanatory machine learning models, Xue et al. [449] explored the main determinants of phase transition temperature of shape memory alloys. By comparing the prediction errors of different machine learning models in the training set, the second-order polynomial model with the characteristics of doping element electronegativity, valence electron concentration, d-electron orbital radius, etc. as input is selected, and the possible mechanism of the dependence of transition temperature on these features is discussed based on the Landau phenomenological model. Esterhuizen et al. [450] developed a theory-guided machine learning (ML) method, which uses an interpretable ML model called generalized additive model (iGAM model) to discover predictive structure-property models that can quantify and explain the relationship between the chemisorption strength and the geometric structure of an adsorption site. A novel deep learning method was developed by Yang et al. [451] to directly simulate the highly nonlinear mapping between multiaxial fatigue life and complex load conditions. This method introduces an advanced deep learning mechanism called self-attention mechanism to describe the effects of varying temperature and complex loading history on fatigue life.

Post-hoc interpretability refers to the development of interpretability

techniques to explain the trained machine learning model. According to the difference between the interpretation target and the interpretation object, post-hoc interpretability can be divided into global interpretability and local interpretability [452]. Global interpretability is designed to help people understand the overall logic behind the complex model and the internal working mechanism, and local interpretability is designed to help people understand the decision process and basis of the machine learning model for each input sample. The classic global interpretation methods include permutation importance [453], feature importance [454], partial dependence plot [455], global surrogate models [456], sensitivity analysis [457], accumulated local effect plots [458], feature interaction (H-statistic) [459], prototypes and criticisms [460], and functional decomposition [461]. For example, Ruiz et al. [462] conducted 529 rotating bending fatigue tests on steel suspension spring bars manufactured with different combinations of manufacturing parameters. A reliable regression model based on the Gradient Boosting algorithm was obtained. The model is explained by permutation importance algorithm, and the correlation between temperature and fatigue life in tempering treatment after quenching is revealed. Li et al. [463] improved the traditional genetic algorithm as a feature selection method for the hardness prediction of high entropy alloys. The gene manipulation and feature importance were introduced into the improved genetic algorithm to make it easier to understand. Deng et al. [464] proposed an interpretable hybrid machine learning (ML) model, which uses polarization ion parameters of molecular dynamics simulation and Particle Swarm Optimization-Random Forest (PSO-RF) model accurately predict the dielectric constant. The PSO-RF model is interpreted by feature importance and partial dependence plots. Surrogate models explain those more complex models through some simple explanatory models, such as linear regression and decision trees, which are often used due to their simple interpretation. Based on the experimental data of synthesis reaction of templated vanadium selenites materials, Raccuglia et al. [465] trained the support vector machine model to predict whether the hybrid materials can be successfully synthesized. The results showed that the accuracy of the model was 89%. Then, they trained the decision tree model with the prediction data of the support vector machine model, established an interpretable if-then criterion, analyzed the thresholds of reactants and reaction conditions, and obtained a new process for the synthesis of this kind of hybrid materials. Bakr et al. [466] adopted the machine learning models trained on large experimental datasets to predict the phase and hardness of high entropy alloy (HEA). In addition, sensitivity analysis suggest that HEAs hardness is affected the most by Cu, Ti, Co, and Ni concentration (respectively). Different from the global interpretability, the local interpretability of the model is guided by the input sample, which can usually be realized by analyzing the contribution of each dimension of the input sample to the final decision result of the model. Classical local interpretation methods include individual conditional expectation curves [467], local surrogate models (LIME) [468], SHAP [469], counterfactual explanations [470] and scoped rules (anchors) [471]. For example, Shi et al. [472] employed machine learning approach to establish a rapid prediction model for shielding effectiveness of polymer nanocomposites/carbon-based fillers, and analyzed the critical rules and factors in material design. In addition, the importance of features is analyzed through the importance ranking of variables, and the influence of key features on electromagnetic interference was investigated through individual conational expectation plots and partial dependence. Park et al. [473] discovered new aluminum (AL) alloys with the assistance of artificial intelligence (A.I.) to improve mechanical properties. Bayesian hyperparameter optimization algorithm was used to predict 7xxx series aluminum alloys, and a high prediction rate was achieved. Under the guidance of the recommended algorithm based on A.I., a new aluminum alloy with excellent combination of strength and ductility was designed. In addition, the relationship between input and output parameters was revealed with the assistance of LIME. Yang et al. [474] proposed a machine learning based alloy design system (MADS) to

facilitate the rational design of HEAs with enhanced hardness. In addition, in order to improve the interpretability of the model, the Shapley additive explanation (SHAP) was introduced, which manifests that valance electron concentration plays an important role in the prediction for hardness. Braun et al. [475] adopted machine learning techniques to predict number of failure cycles and failure locations of fatigue tests performed on small-scale butt-welded joint specimens. Furthermore, the SHapley Additive exPlanations (SHAP) framework was used to identify influential features and their interactions. Finally, the model interpretation was linked back to the domain knowledge.

5. Conclusion and prospect

The application of machine learning in the field of materials science and engineering has brought a completely different research perspective from the traditional experiment and simulation methods and showed great potential due to its flexibility, accuracy and good generalization ability in dealing with complex systems. The application of machine learning is highly interdisciplinary, and there are many application objects of machine learning in materials science. Due to space limitations, this paper focuses on the application of machine learning in metal materials, and summarizes the research progress of machine learning in material design and optimization. A variety of related studies have shown that machine learning can be used as a fast and accurate tool in the field of materials science. Further development of theories and methods will lead to a more in-depth and extensive application of machine learning in the field of material science.

Looking forward to the future, the accumulation, standardization and sharing of material data will promote the application of machine learning in the field of materials, and the construction of large-scale databases will promote material research and development into the era of big data. Therefore, studying the application of machine learning in material science under the background of big data is an important research direction in the future. In addition, in view of the "no free lunch" theorem in machine learning theory, the development of ensemble learning or universal algorithms to meet the needs of different materials design will promote the wide application of machine learning in the field of materials; Developing machine learning strategies under the background of "small data samples" + "large component space" will overcome the problems of sparse experimental data, difficult acquisition and slow speed in the field of materials science at this stage, and help to improve the learning efficiency and accuracy under limited data; The development of feature tools and automated machine learning (AutoML) framework will lower the entry threshold for machine learning and help materials domain experts who are not familiar with data science to quickly extract value from their data with little effort, time and cost; Developing inverse design strategy of design on demand, which can combine the specific properties requirements of materials, give full play to the initiative of using ML, and ultimately accelerate the development of new materials; Develop interpretable machine learning algorithms and models, which help to deeply understand the mapping relationship between material data and machine learning models to eliminate inconsistencies between the interpretation results and the real behavior of the model, thereby ensuring the reliability and security of the interpretation results.

In brief, material research and development based on machine learning is a new paradigm of material research, which will accelerate the development efficiency of materials and reduce costs. This new research interests combines material science, data science, computer science and artificial intelligence. It is a multidisciplinary research field that requires experts in the material field and computer field to collaborate with each other to further promote its rapid development.

CRedit authorship contribution statement

Jiheng Fang: Conceptualization, Methodology, Software,

Investigation, Formal analysis, Writing – original draft. **Ming Xie**: Data curation, Writing – original draft. **Xingqun He**: Visualization, Investigation. **Jiming Zhang**: Resources, Supervision. **Jieqiong Hu**: Software, Validation. **Yongtai Chen**: Visualization, Writing – review & editing. **Youcai Yang**: Visualization, Writing – review & editing. **Qinglin Jin**: Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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